

Choice, Competition, and Risk at Work: Essays on the Economics of Workers' Compensation Insurance

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Preamble

I am eager to present the projects I have spent much of my PhD working on. With the goal in mind of going on the market in Fall 2024 and defending the final dissertation in Spring/Summer 2025, this proposal outlines - with in-development versions - two papers and ideas for a third. The ideas for the third chapter are all based on data already in-hand, or very close to being in-hand, and they represent actionable items that could lead to future, fuller projects and are doable by next spring. I am eager for your feedback and look forward to our discussion, both now and ongoing. The first two chapters are organized by how far along they are, but not necessarily by importance or in order of which one should be my job market paper (JMP).

What parts of the US workers' compensation system work best from the perspective of employers, employees, and insurers? This question is the guiding hand of the dissertation and, hopefully, the agenda that can be built for some part of the tenure track. Of course, this question is too big for any one paper. Yet, that is ideal given this dissertation, and my research agenda moving forward, will focus on individual components of this complex, state-based system to help inform policymakers.

Throughout our discussion of this proposal, I would like to achieve two things, in addition to taking all of your (predictably wonderful) feedback: 1) have an idea as to which paper - Chapter I or II - would best serve as my JMP (taking into consideration the need to maximize across markets - risk and insurance, public policy, economics, etc.) and 2) decide on an avenue for the third paper that will work best to be in-development at the time of the final dissertation defense. I thank you in advance for your time, advice, and for providing such a great environment in which to get a PhD.

Below, the Preamble presents chapter topics, ideas, and current statuses. Then, the formal packet commences with the current version of the paper making up Chapter I, before moving to Chapter II and eventually closing with the list of ideas from which Chapter III will develop.

Chapter I

What is the causal effect of introducing a government-organized competitor into the private market? I use the workers' compensation market to provide the first evidence as to this ques-

tion in the US insurance context. I take advantage of variation in state-based insurance markets, where approaches range from government monopolization to full-fledged private market competition to the government joining as an active competitor (i.e., as a “public option”). Using data from the National Association of Insurance Commissioners (NAIC), the National Academy of Social Insurance, Census’ County Business Patterns series, and various supplemental requests directly from a public option and a rating agency, I focus on the period of the early 1990s wherein eight states create new public options between 1992-1997, diverging from their previously private-only marketplaces.

I document several empirical changes in workers’ compensation markets after the creation of a public option. Despite elevated levels of concentration in new public option states before formation, via increasing market shares of a few private insurers, the introduction of a public option leads to substantially more concentrated markets in both the short and long run. As one measure of an insurance company’s operations, I also study changes to the loss ratio, which is the percentage of a firm’s premiums received that are paid out in losses to policyholders. A loss ratio above one indicates that an insurer is paying out more money in losses than it is earning in premium, even before accounting for expenses. Before public options were created, loss ratios were elevated in eventually-treated states relative to control states, with statewide average loss ratios exceeding one for several years, which is unsustainable. After the public option is created, loss ratios fell significantly, indicating improved performance and profitability across insurers within a state.

There are several explanations for improvements in the profitability of workers’ compensation insurance markets at this time, given that we should not expect that the public option is able to deny claims at any greater rates than its private counterparts. One explanation is that benefits, which are prescribed by statute in each state, were decreased at the same time as part of a larger reform period. On its face, though, there is little reason to expect that benefits themselves reduce loss ratios, as both elements of the ratio stand to decrease, albeit on different time lags. Instead, the argument must be that premiums did not respond to benefits decreases (costs) as much as losses. I show that, while benefits paid decreased by around \$0.01 per \$1 of payroll after public options were

created, market prices decreased by only \$0.005 or so. When you look only at private insurers, however, the price decreases are larger (and more persistent). Over time, the private insurer price reduction remains larger than the market price reduction altogether. This helps explain the market improvement as a function of public options gaining market share while avoiding lower prices. The loss ratio results, for example, bear this out: loss ratio reductions after a public option joins the marketplace would be smaller if not for the public option's performance. Analysis of other mechanisms indicate that changes in industry mix and the residual market responsibility of the public option do not explain the results.

Next Steps: As you will see when you read the full paper, a lingering question remains in this context: how do public options take market share in certain states if it is not via low prices? An exploration into this dynamic would be an ideal way to conclude the paper, and heterogeneity among the states will be easier to study given I now have all eight public options' financial data (only six of the eight were in-hand prior to late April 2024). It may also be optimal to ask a bit about the private market impacts from the public option by providing some descriptive evidence as to whose market share shrinks after a public option enters: single-line insurers (those who sell only workers' compensation) and/or multi-line insurers (those who sell other types of insurance, too, like commercial liability or auto). This is relevant given many companies seek to bundle their insurance products (i.e., buy workers' compensation along with other commercial insurance) so that a public option that gains market share in only workers' compensation may have spillovers into other insurance markets.

Chapter II

Do employees benefit from an optional workers' compensation coverage setting, one where employers choose whether to be covered and employees choose where to work? I use data from several unique open records requests in Texas, the state with the only robust opt-out path for workers' compensation in the US, to provide the first evidence of losing workers' compensation insurance coverage on employee outcomes. These data come from statutorily required filings by employers who opt to be a subscriber (by purchasing insurance or setting up a self-insurance

arrangement approved by the state) or a non-subscriber. Subscriber status is important because subscribers (insured or self-insured) face no risk of an employee pursuing a tort claim, given the workers' compensation system precludes other avenues for an injured worker. Non-subscribers may be sued in court, and may or may not have more informal injury plans governed by the Employee Retirement Income Security Act of 1974 (ERISA, for short). In these data, I can see firms and individual establishments, with county and industry information, for the 2010-2022 period.

As a first question, I ask if the uncovered rate (percent of total establishments making an annual uncovered filing with the state, i.e. the rate of non-subscription) in a given industry-county pairing is related to wages, both in terms of total wages paid, total taxable wages paid, average weekly wages, and average annual wages. The establishment and wage data come from the Quarterly Census of Employment and Wages (QCEW) produced by the Bureau of Labor Statistics (BLS). The results indicate that as the uncovered rate increases, wages also decrease. This result is perplexing, as compensating differential theory predicts a positive relationship, if employees are aware that losing workers' compensation coverage ought to be accompanied by an increased direct wage. The results are consistent when using a binary treatment where the continuous uncovered rate is replaced with an indicator variable equal to one when an industry-county observation has experienced an increased uncovered rate from one year to the next.

This analysis alone, even with industry-by-county and year fixed effects, is plausibly endogenous. Unobserved injury rates at the firm level likely influence wages and the uncovered decision. As a result, I suggest an instrument: whether or not a given county experienced a change in judge (or judge's party, with partisan judicial elections in Texas) at the district court level, where personal injury cases (what an employee would file against an uncovered employer) would be filed. This data was hand-recorded, covering all statewide district judge elections in Texas between 2002-2020 (thus covering incumbent/non-incumbent term histories over the 2010-2022 period), via the Texas Secretary of State's election results archive. This is a compelling instrument because it is empirically relevant to uncovered rates, and it is plausibly exogenous – conditional on county size and business settings (gathered from the Census and Bureau of Economic Analysis, respectively)

– in that the only way a judge election would alter the wage is by directly affecting the tradeoff an employer is making when considering to be uncovered and, thus, open to liability risk.

Next Steps: As a first step, I want to provide more descriptive evidence surrounding the first stage, and there is likely to be great value in presenting more information on the relationship between being a non-subscriber and other information at the industry and/or county level, including injuries, job transfers, lawsuits, and personal injury law firm advertising. This will better support a stronger argument for two-stage least squares estimation. I also intend to work on this project, especially the injury component, over the summer as a Fellow in the Chief Evaluation Office at the Department of Labor (DOL). To do so, I will make use of injury-level data among insured subscriber firms from an additional open records request in Texas, which presents data over the 2015-2021 period. The DOL is aware, and supportive, of the fact that my exploration into the injury data is a part of my dissertation.

Chapter III

The same insurer-level and injury-level data described above enables me to pursue several different projects as a third chapter. Below, I briefly discuss these ideas (in no particular order, using numbers only for ease of reference). For completeness, I repeat this page as the third chapter at the end of this proposal, which includes some additional discussion.

1. Solo: Before mid-2017, all workers' compensation fraud concerns were handled by a specialized department – with criminal referral powers – housed within the state's public option, Texas Mutual Insurance. After concerns about the powers of that fraud unit and pressures from other insurers, the state created a government agency to study fraud and replace this arrangement. Using fraud indicators provided by insurers in the Texas injury data, I would like to study the prevalence of fraud before and after, which not only answers a public-versus-private provision of fraud investigations efficiency-related question but could also lend credence to using NAIC data to study the performance of the public option before and after this change relative to its private market competitors with detailed expense information.

2. Solo: I am interested in using a change in rules associated with compound drug pre-authorizations to deal with prescription costs and opioids in treatment. Texas became concerned about costs and a loophole in their new formulary, which allows drugs made up of ingredients on the formulary individually to be approved together as a compounds. These compound drugs are a concern for medical reasons, as they not approved by the FDA, and for cost reasons, given they are expensive and relevant for potentially addictive pain treatment.
3. Joint with Ty Leverty: Many public options are exempt from federal taxation. There is no quantification of this tax advantage, however, and this topic is relevant to the broader question of ways in which workers' compensation systems are optimal or sub-optimal for various stakeholders. This project can make use of several state changes in the federal tax exemption (Maine and Utah), along with a few public option privatizations (Arizona and upcoming in Missouri), to study the impact of the federal tax exemption on competition and prices, for example.
4. Joint with Honglin Li: Workers' compensation is one of the more highly experience-rated lines of insurance in the US. Experience rating is often discussed as a way to solve various informationally-driven market failures. Using reported experience rate modifiers in the injury data from Texas, we propose to study the effect on injuries and claims from 1) changing the size of a claim included in a modifier calculation, and 2) the use of a cutoff in experience rating for small firms. With data available between 2015 and 2021, and these rule changes taking effect in mid-2015 and into 2016, observable policy effectiveness dates allow us to study temporal changes.
5. Joint with Honglin Li: In 2019, the state of Texas stopped allowing negotiation of experience rate modifiers. This is an important change because some contractors may lose out on bids if their experience rate modifier is above 1 (indicating an injury experience more than their industry's average). We plan to use this change to think about bargaining power with reported experience rate modifiers and the linked, detailed injury-level data.

Chapter I - Government Involvement in a Mandated Insurance Market: The Case of Workers' Compensation Insurance*

Abstract

Nearly every state requires workers' compensation insurance, which provides workers with financial protection for injuries at work. With each state in the US regulating its own insurance market, the organization of marketplaces differs widely. In some states, there is open competition among private insurers. In other states, there is a government-run monopoly. Yet, in other states, a government-run insurer ("public option") competes with private insurers. There are also differences over time as public options are created or eliminated. I exploit this variation to estimate the causal effects of government involvement in a mandated insurance market, which lends insight to adopting a public option in health insurance. I document several empirical effects of a public option. The marketplace becomes more concentrated with public options gaining market share while underwriting profitability improves, driven by reductions in price that are smaller than reductions in statutory benefit levels. Public options appear to gain market significant market share while not charging lower prices than their private counterparts. I also discuss several potential mechanisms, such as residual market responsibilities, industry mix changes, and firm entry behavior.

Keywords: workers' compensation, public option, insurance markets, social insurance

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1 Introduction

How governments regulate markets takes many forms, from price controls to quality standards to direct government competition with the private market. This competition mechanism is usually a “public option,” whereby the government provides people with a publicly-funded option to private alternatives (like public libraries, pools, and schools). The public option receives ample attention in the health insurance setting. In the 2020 US Presidential election, nearly every Democratic candidate proposed a public option to reform health care (Abrams, 2019). Despite the attention, there is no existing empirical evidence of the effect of a public option on US insurance markets.

To gain insight into the impact of a public option on US insurance markets, I study workers’ compensation insurance. Health and workers’ compensation insurance have comparable risk protection. Workers’ compensation insurance pays benefits when a worker is injured, providing medical coverage and a portion of lost income. And like most health insurance, individuals get workers’ compensation coverage from their employers. Yet, unlike health insurance, many states have a public option for workers’ compensation insurance provision with a notable market presence, paying nearly \$5 billion in workers’ compensation benefits in 2021 (Welch et al., 2024). Using detailed, insurer-level data from the National Association of Insurance Commissioners (NAIC), supplemented with state-level workers’ compensation benefits and payroll information coming from the US Census Bureau’s County Business Patterns data and annual reports from both the *Social Security Bulletin* and the National Academy of Social Insurance (which are gathered for the 1970-2020 period), I study the causal effect of public option formation on the market, focusing on competition between insurers, insurer underwriting profitability (i.e., how much an insurer pays out in losses relative to what it earns in premiums), and a measure of workers’ compensation insurance prices per dollar of payroll.

There is incredible variation in the structure of workers’ compensation insurance markets across the US. In some states, the government is the only seller; in others, only privately-

owned insurers sell coverage. Yet, in other states, there is a mixture of these two approaches, where the public option (also known as a “state fund”) competes with private insurers. Over time, many states have changed their market involvement via the public option, by either creating one for the first time or privatizing a currently-operating one. To foster this analysis, I have documented, from 1970 forward, the workers’ compensation market status in each state along with details about the public options and their founding statutes, which allows for an understanding of a given public option’s federal tax-exempt status and residual market responsibilities (i.e., if guaranteed issue serves as a policy mandate).

This study seeks to answer several research questions by exploiting these market differences. The paper proceeds in the order of the policy-making process – dealing with motivations, then effects of policy changes, and finally understanding the mechanisms behind what is observed empirically. First, what did workers’ compensation markets look like before new public options were formed in the early 1990s? In particular, it is relevant to ask about several dynamics: how insurers were performing, how many may have been leaving or entering, along with ensuing competition effects, and what prices employers were facing to buy standardized coverage.

Second, what is the causal effect of public option creation in terms of those same workers’ compensation insurance market dynamics? States have been addressing the roles of their public options for quite some time. After a flurry of public options were created in the early and mid-1990s (in Kentucky, Louisiana, Maine, Hawaii, Missouri, New Mexico, Rhode Island, and Texas), several states have relaxed the role of the public option since (Nevada, West Virginia, Minnesota, Arizona, Maine, and Utah), including privatizing it entirely. In fact, Missouri will do so by 2025. The effects of the public option could also be felt outside of the workers’ compensation insurance market. For example, competition changes in the workers’ compensation market may exhibit spillovers to other commonly-bundled commercial insurance products, such as commercial auto and general liability lines.

And third, how does the public option affect prices of workers’ compensation insurance?

Below, I propose a new standardized price measure in workers' compensation insurance, which addresses issues in the insurance economics literature of visualizing premiums but not price as premiums are a combination of price and quantity. This price measure can then be compared to changes in the amounts of benefits actually paid to workers, as provided by statute, to understand how gaps between the two may tell a story about the dynamic of the marketplace from the perspectives of several stakeholders.

There are many plausible motivations to form a public option, generally related to market failures. Government involvement in insurance markets began with workers' compensation (Greene, 1976), and the rationale typically given – and produced by Greene (1976) – is that the government ought to be involved in insurance because of residual market needs (i.e., the private market is not making coverage available), convenience, compulsory purchase, efficiency, or collateral social purposes. In the workers' compensation argument for competitive state funds that exist as a public option, the reasoning is typically efficiency related or due to affordability and availability issues (Thomason et al., 2001).

Using insurer-level data from 1988-2020, I first study concentration in workers' compensation insurance markets. With states that maintained consistently competitive markets (i.e., no public option) as the counterfactual, public option states experienced rising concentration levels (as measured by the Herfindahl-Hirschman Index, or HHI, which increases when more insurance is provided by just a few insurers) before public option creation. After the public option begins operating, those states see an even larger jump in concentration, ultimately never returning to the levels of the competitive counterfactual. Relative to the gap in concentration levels between public option states and completely private states just before public option create, the results indicate a near doubling of the HHI, where most new public options began to take significant market share.

I also show that insurer underwriting profitability dramatically improved after the public options commenced operation. As measured by the statewide average loss ratio in the workers' compensation line (which measures the insurers losses - from current claims and expected

developments of those claims over time - relative to premiums), loss ratios were significantly higher (more than ten percentage points higher) in public option states before creating the public option. In fact, they were above one for several years, an unsustainable proposition for insurers who are operating at a loss before including their operating expenses. After the public options joined the market, loss ratios immediately corrected back to the competitive counterfactual and, in some years, public option states experienced better performance than their competitive counterparts. Some of these loss ratio results, however, can be explained by drops in legislated benefit levels (roughly \$0.01 per dollar of payroll) in public option states that were larger than the response in prices (roughly \$0.005 per dollar of payroll). This is especially true for the public option, which generally experienced lower loss ratios over time than their private competitors.

This study contributes to the literature in a few important ways. To start, this is the first study to empirically estimate the effect of a public option in US insurance markets. Public options have been widely studied in other US contexts ([Hart et al., 1997](#); [Amirkhanyan et al., 2008](#); [Caskey, 2010](#); [Levitin and Wachter, 2013](#); [Dinerstein and Smith, 2021](#)) and internationally ([Polyakova, 2016](#); [Bergman et al., 2016](#); [Atal et al., 2024](#); [Knutsson and Tyrefors, 2022](#)). In the US non-health context, one of the most common areas to study a public option is in education, where [Dinerstein and Smith \(2021\)](#) provide recent evidence in this arena and review prior work. Other examples exist, too. [Caskey \(2010\)](#) analyzes off-street parking in Philadelphia, [Levitin and Wachter \(2013\)](#) discuss household finance and mortgages, and [Hart et al. \(1997\)](#) discuss public provision of prisons and garbage collection, among others. [Amirkhanyan et al. \(2008\)](#) study nursing homes in the US, comparing for-profit provision to non-profit and public provision, and ultimately show evidence of for-profit inferiority.

In the international health insurance context, [Polyakova \(2016\)](#) studies the German setting, for instance, finding that private insurers don't seem to "cream skim" good risks systematically. [Atal et al. \(2024\)](#) study the Chilean public pharmacy system, finding that private

pharmacy prices increased upon public pharmacy formation, evidence of private pharmacies retaining less cost-sensitive consumers. [Knutsson and Tyrefors \(2022\)](#) study competition between public and private ambulances in Sweden, finding private advantages in some measures (costs, response time) and disadvantages in others (mortality). Also in Sweden, [Bergman et al. \(2016\)](#) study the privatization (note that this is slightly different than studying the creation of a public option) of elderly care centers, finding private firms improve mortality.

In US health insurance markets, researchers have focused on understanding what a public option could do ([Cebul et al., 2011](#); [Miller and Yeo, 2019](#); [Craig, 2022](#)), including using “Medicare-for-All” ([Shepard et al., 2020](#)) or focusing on certain state-based arrangements that look like public options but function in (importantly) different ways. For instance, Washington recently implemented an approach where private insurers could sell a “public option plan” with regulated features ([Sen et al., 2022](#)). Colorado took a similar approach with running public option plans through private insurers but mandated that private carriers offer one ([Kumar and Adashi, 2023](#)). Nevada is set to implement a similar approach by 2026. All fall well short of the traditional public option design present in workers’ compensation and as has been called for in the national health insurance setting. There are other insurance arrangements that have public options but are yet to be studied (to the best of my knowledge), namely mandatory non-occupational disability insurance ([Greene, 1976](#)). While [Greene \(1976\)](#) notes many other examples of government involvement in insurance markets, it is typically not in competition with the private market and is offered in partnership with private insurers.¹

This study also contributes to the literature on spillovers of insurance regulation, both across states and across lines of business. The most direct connection made is usually between health insurance and workers’ compensation, and authors have studied the spillovers of health regulation (i.e., the Affordable Care Act) on workers’ compensation insurer perfor-

¹There has been ample attention paid to public versus private provision of insurance in terms of traditional Medicare (public provision) and Medicare advantage (private provision) plans ([Curto et al., 2019](#)). Importantly, that setting differs from the pure public option, given the government provides set payment amounts to private carriers for each enrollee. My analysis below focuses on public options in their purest form.

mance (Baggett et al., 2020) and claims (Dillender, 2015). Studying spillovers across lines of insurance, as I do in this paper, is slightly different from the recent work on spillovers between states based on regulatory frictions (Oh et al., 2022). Scope economies in insurance, which relate to how insurers may operate across lines of insurance, has been recently analyzed in the health insurance setting (Born et al., 2023). In that paper, the authors provide evidence against scope economies (and for strategic focuses) and discuss the prior work in other types of insurance.

Naturally, this study also contributes to the broad literature on workers’ compensation insurance. In light of recent work outlining the lack of market failures to inspire mandates for workers’ compensation insurance coverage (Cabral et al., 2022), policymakers may consider turning to the design of Texas’ system, which has no coverage mandate and a public option. Importantly for state-based policymakers considering these changes, understanding what the impact of the public option is on the workers’ compensation market helps isolate one key policy lever. This is especially relevant with Missouri passing a bill in 2023 that will privatize its public option by 2025, which could encourage other states to reconsider changes to their setting in a market like workers’ compensation where state action has come in bunches.² In the workers’ compensation literature, this would be the first study to apply robust causal inference strategies to learn about public options, not just in the years of creation but over the long term, with more than 30 years of detailed, insurer-level data. There has been ample research on workers’ compensation insurance markets over time, but it has solely focused on the markets without public options (for example: Carroll (1993); Danzon and Harrington (1998, 2001); Harrington and Danzon (2000)), the cost efficiency differences between private and public insurers (for example: Butler and Worrall (1986); Thomason et al. (2001)), or the moral hazard effects of benefits themselves (see Meyer et al. (1995) for just one example).

Given the rich history of workers’ compensation and the state-based insurance markets, Section 2 describes the institutional background. Section 3 describes the data, while Section

²Missouri SB 101 was signed into law in July 2023: https://senate.mo.gov/23info/BTS_Web/Bill.aspx?SessionType=R&BillID=44649.

4 presents the empirical methodology. Sections 5, 6, and 7 present results, discuss prices and mechanisms, and conclude, respectively.

2 Institutional Background

In the early 1900s, the fight for workers' compensation insurance was just getting underway. With no federal movement, it largely became a state issue. Several debates surfaced about how to best operationalize workers' compensation programs, with a more niche debate erupting in many states over how to provide workers' compensation insurance. Meaning, should it be provided with government involvement, either in competition with the private market or in a monopoly arrangement? This debate featured disagreements between organized labor (desiring state monopolies), insurers (desiring no state involvement), and cautious employers (not wanting poor state fund control to lead to higher costs and worried about adverse selection) that led to heterogeneity across states that has persisted to this day (Clayman, 1959; Weinstein, 1967; Fishback and Kantor, 1996, 1998).

Before discussing the state variation in workers' compensation insurance market formation in detail below, it is useful to provide a brief idea of what the system does for workers. Workers' compensation provides almost guaranteed payment to employees for injuries at work (thus, it is an occupational disability program).³ Yet, it is also the exclusive remedy available to employees (employees cannot sue their employers). This is an example of a strict liability system. As long as an employee injury satisfies a few conditions (harm caused by an accident arising out of, and in the course of, employment), they will be paid for some lost income and all relevant medical expenses. Without a workers' compensation insurance program, typical negligence rules would apply, whereby the employer and employee must resolve the issue in court (or settle out of court), with more risk – both in who wins and how much payment is warranted. Coverage for workers' compensation is provided by the employer for its employees; workers' compensation is not a product available for employees

³Though, this does not mean non-occupational injuries or illnesses do not find their ways into workers' compensation claims. Smith (1990) documents the prevalence of increased claims on Mondays that depict evidence of concealment and delayed treatment.

to purchase directly. Benefits are prescribed by the state, so each firm is purchasing the same amount of coverage per employee in this setting.

2.1 Workers’ Compensation Insurance Marketplaces

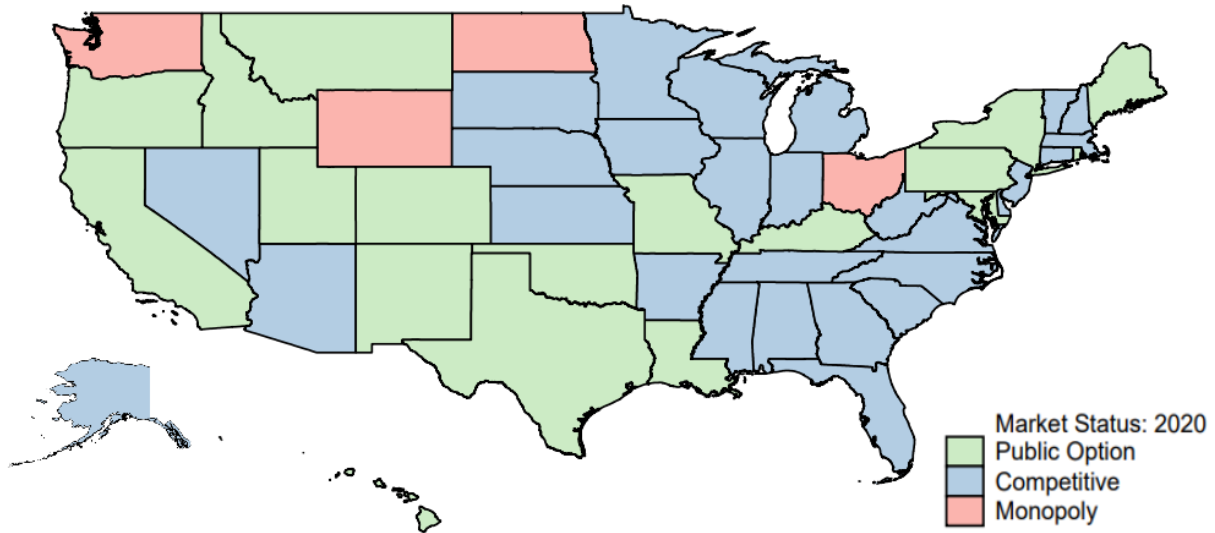
The competitive structure of the workers’ compensation insurance marketplace varies from state to state (and over time, as will be described below). Categorizing the marketplaces is a difficult endeavour, as there are many ways (some being far beyond the scope of this paper) in which even seemingly similar marketplaces differ. The goal of this paper - to study the entrance of “public options” - requires, at a minimum, broad categorizations of marketplaces.

To broadly categorize the state⁴ marketplaces, I start with thinking about how employers are permitted to provide workers’ compensation coverage to their employees, while retaining the exclusive remedy nature: either through buying insurance or arranging self-insurance. Within the insurance path, there are three broad categories that define the US workers’ compensation insurance market: 1) monopoly states, 2) competitive states, and 3) public option states. Figure 1 provides the state of the US marketplace in 2020 (or even through 2023, given there have been no changes since 2020).

In monopoly states, of which there are four, the only source of coverage is via the government. There may or may not be an option for employers to become certified self-insurers (North Dakota and Wyoming do not allow self-insurance). In competitive states, there is no government insurer presence and the only source of coverage for employers comes via the private market. In these settings, the government usually sets up a residual market mechanism (or residual pool) which reinforces the idea of compulsory coverage for those employers who cannot get coverage on their own. Finally, there are public option states. In these settings, the marketplace is a competitive one where an active competitor is the public option insurer, which was created by the government and typically operates like a

⁴Here and throughout the paper, I refer to states and jurisdictions interchangeably and include the District of Columbia (DC) in those discussions - which is why I reference 51 “states” or jurisdictions. I do not reference any US territories in this paper, though they each have their own market structure.

Figure 1: Workers' Compensation Market Structure in 2020



Notes: This figure displays broad workers' compensation insurance market structure as of 2020. Sources: National Academy of Social Insurance annual reports and author's own statutory & policy research. Sources for this statutory and policy research are compiled and available upon request.

mutual insurance company. Both competitive and public option structures also allow for self-insurance arrangements.

There is additional variation in public option states, most notably related to residual market responsibilities and federal tax payments. Four of the eighteen states with a public option - Idaho, Missouri, New Mexico, and Oregon - do not require that the public option accept all risks. Meaning, the public option does not serve as the residual market mechanism. In the other fourteen states with public options, the public option does serve as the residual market.⁵ Within the fourteen public options that are also the residual market mechanism, two of those states - Maine and Utah - are no longer classified as public option states strictly by the National Academy of Social Insurance, for example, because they have become more privatized (namely with respect to their obligation to pay federal taxes, which other public options do not need to do). In the map above, I have kept those two states as public option

⁵Though discussions with an executive at a public option insurer suggest that there is very little in the way of risks being classified as "residual risks," at least today.

states for simplicity. The forthcoming analysis does not focus on Utah and will only assess Maine’s public option when it was still exempt from federal taxation.

This paper does not account for all sources of variation across states. Rather, the aim here is to first account for the role of the public option’s creation and then to ask about ways in which the causal effect may be heterogeneous across the reform spectrum. In addition, coverage is mandated in nearly every state, except Texas.⁶ Understanding how that may play into the public option’s long term success is of interest, especially with ample state activity within the last decade being focused on duplicating Texas’ design of its workers’ compensation program, i.e. not mandating coverage and using a public option (Cabral et al., 2022).

2.2 Changes to Market Structures Over Time

Figure 1 provides a cross-section of state workers’ compensation insurance markets, but the motivation of this paper comes not only from state differences statically, but *changes* in these differences over time. Figure 2 provides a mapping of state marketplaces from 1980 through 2020. While this information was tracked between 1970-2020, there were no changes between 1970-1980, so I omit the 1970s for space efficiency.

The color discontinuities in this graphic depict structural changes to the marketplace. For instance, Nevada went from a monopoly state (in red) to a competitive state (in blue) in 1999. The same happened in West Virginia in 2007. This graphic not only helps identify the time of changes, but it highlights just how much variation there is across the US, by state and over time. For methodological reasons, it also presents reassurance that there were many states not making changes over this period.

Beyond the privatization of monopolized public options in Nevada and West Virginia, there were nine states that created a public option over this time period: Minnesota (1984), Louisiana (1992), New Mexico (1992), Rhode Island (1992), Texas (1992), Maine (1993),

⁶Wyoming and South Dakota also may allow non-hazardous employers to opt out, but there is little evidence that the opt-out is used very often. In Texas, however, a third of employers opted out of the workers’ compensation system in 2020, which included 20% of the state’s employees (DWC, 2020).

Figure 2: Workers' Compensation Market Structure Changes, 1980-2020



Notes: This figure displays broad workers' compensation insurance market structures from 1980 through 2020. Blue corresponds to competitive states, while red corresponds to monopoly states and green to public option states. Sources: National Academy of Social Insurance annual reports and author's own statutory & policy research. Sources for this statutory and policy research are compiled and available upon request.

Kentucky (1995), Missouri (1995), and Hawaii (1997). While Minnesota had completely privatized its public option by 2008, the other eight states continue to have public options, though with variation across them as discussed above. Minnesota was not the only state to privatize a state fund over this time period, however. Arizona (2013) and Michigan (1994) have also done so, and Missouri plans to do the same by 2025.

3 Data

This paper employs data from numerous sources that can achieve a two-pronged goal. First, I use detailed, insurer-level data at the state level to assess the causal effect of public option formation on market structure, competition, and performance. Then, I use a long panel of state payroll figures and payer-level information on workers' compensation benefits paid to complete the picture and explore mechanisms.

3.1 Insurer Data by State, Line of Business, and Year

As is commonplace in the insurance economics literature, I utilize data from the National Association of Insurance Commissioners (NAIC) to study the impacts of public option creation on insurance markets. These data exist between 1984-2020 and cover the universe of

insurers over that time period, though with better representation in 1988 and later. Additionally, these data allow for visualization of each line of insurance a firm writes in each state in which they operate (workers' compensation, personal auto, home, commercial liability, etc.). Thus, it is possible to directly observe a firm's workers' compensation insurance information in each state and year, providing insurer-state-line-year observations.

In the workers' compensation insurance setting, there are a few important advantages of the NAIC data. These advantages, of course, are beyond the distinct benefit of the NAIC data in providing the most comprehensive set of insurer-level information so far back in time. First, the public options that were created in the early 1990s - the ones this paper studies empirically - all appear in the NAIC data when they are first created, which aligns perfectly with the statutory development of creation dates (as shown in the earlier figures).⁷ Coverage of longer-existing public options, such as in California, or state-run monopolies is not as robust. Second, the NAIC data allow for the building of a picture of other lines of business (like commercial auto and general liability) that are commonly bundled with workers' compensation insurance for an insurer's commercial clientele.

3.2 Benefits Paid Data: 1970-2020

The only way to piece together payer-specific information on workers' compensation benefits over the same period for which statutory data was gathered (1970-2020) is to combine data from three sources: annual *Social Security Bulletin* reports, annual National Academy of Social Insurance reports and data releases, and the Census' County Business Patterns (CBP) data series. Each of these is discussed in detail below.

3.2.1 *Social Security Bulletin* and National Academy of Social Insurance Data: Workers' Compensation Benefits by Payer

Data on workers' compensation benefits payments come from two primary sources: 1) *Social Security Bulletin* (SSB) publications between 1970-1993 and 2) National Academy

⁷Texas and Louisiana appear after they were statutorily created, but I was able to find the necessary information to fill in the gap from gracious industry contacts.

of Social Insurance annual reports or data releases between 1994-2020. The latter took up reporting the same information as the former once reporting ceased after 1993, so the reports – despite coming from two sources – are intended to be a continuation of the same effort. Both of these sources are largely available publicly.⁸

For each year between 1970-2020 and for each state, I am able to identify the amount of workers’ compensation payments made by 1) private insurance carries, 2) public options (state funds), and 3) self-insured employers. These items are consistent across both sources of reports. The reports note that these include cash and medical benefits paid, and they are taken from state documentation (published and unpublished) and may be estimated where necessary. Generally speaking, these data provide vindication for Figure 2 with seeing public options turn on and off.⁹

3.2.2 County Business Patterns Data: Employment, Payroll, and Firms

A natural next step is to be able to normalize workers’ compensation measures, which seminal work suggests should be done with payroll (Danzon and Harrington, 2001). Annual employment, payroll, and firm (number of reporting units) data, at the industry-state level, from 1970-2020 is gathered courtesy of the Census’ CBP offerings. These data are available from the Census (1986-2020) and National Archives (1970-1985) websites. Prior work has graciously shared some digitized data, thanks to the authors’ manual efforts, from 1971-

⁸The following reports provide the data: Skolnik and Price (1973); Price (1975, 1977, 1979, 1981, 1986); Nelson (1988, 1990, 1992, 1993); Schmulowitz (1995); Mont et al. (2002); Sengupta et al. (2005, 2010, 2013, 2014); Sengupta and Baldwin (2015); Baldwin and McLaren (2016); McLaren and Baldwin (2017); McLaren et al. (2018); Weiss et al. (2019); Murphy et al. (2020, 2021); Murphy and Wolf (2022).

⁹The SSB publications mistakenly suggest public options in New Mexico and Rhode Island did not begin paying when they actually did, so those two states are dropped from the benefits paid estimates because there is no easy way to know how much of the private carrier payments should be from public options. A note must be made about Hawaii, which was cited as having a state fund in 1992 and 1993 despite the state fund not being created until 1997 (Schmulowitz, 1995). This was an error not acknowledged by the SSB publications (as they ended with the 1993 year), given the Hawaii government noted in a separate report that these were from the state Special Compensation Fund (Pang et al., 1993) and the National Academy of Social Insurance did not report Hawaii state fund losses paid between 1994-1996. Following the National Academy of Social Insurance’s process of prorating the special injury fund payments across privately-insured and self-insured losses, I take the (incorrectly reported) state fund losses paid value and add it to privately-insured losses and self-insured losses, based on the percentage of total losses paid by private insurers and self-insured firms.

1973 (Eckert et al., 2022). This paper notes how unique this data is to economic analysis, with almost no work using the data, simply because of the lack of digitization. Between 1975-1985, the digitized data come from another paper by some of the same authors (Eckert et al., 2021).¹⁰ For 1970 and 1974 (National Archives), along with 1986-2020 (Census), I hand-collected the data, just as the authors of the prior papers did. The main advantage of those two papers was that the authors provided complete US files, instead of breaking it out into nine geographic regions (as the National Archives’ website does in the raw form from 1971-1973 and 1975-1985).

One weakness of the CBP data between 1970-1973 is that annual payroll is unavailable. Instead, first quarter, taxable (subject to the Federal Insurance Contributions Act or FICA) payroll is reported. The Census suggests that this first quarter, FICA-taxable payroll number is within 1% of total first quarter payroll, given FICA taxes apply to all earnings except for extremely high earnings due to Social Security income limits (Census, 1988). However, turning first quarter payroll into annual payroll is not as straightforward as multiplying by four. Thus, for these data, I create a state-specific “payroll factor” from 1974-2020 (which I define as total annual payroll divided by first quarter payroll, averaged over the 47 year period). Then, I apply this state-specific factor to each state’s first quarter FICA-taxable payroll in 1970, 1971, 1972, and 1973 to create annual payroll for 1970-1973.

To be clear in summarizing, the CBP data I use include (with my note above about making first quarter payroll into an annual number) the following values from 1970-2020: number of establishments (reporting units), payroll (both annual and first quarter), and employment (measured mid-March). An important assumption for people in the workers’

¹⁰With data from 1971-1973 and 1975, I have to manually sum across all industries or counties within a state. This does not require additional assumptions. The 1975 data is published in combined form by Eckert et al. (2021), but the annual payroll data did not make sense, in that paper nor as published by the National Archives, because total payroll in one industry exceeded the reported total annual payroll in the entire state. Thus, for 1975, I complete this same summation procedure. All other reported totals in that 1975 file are correct – and the error was as simple as, for Maine, reporting total annual payroll as “2102” when it really was “2101529” (supposed to be in thousands, per the data handbook). Thus, it seems as if the total number was really in hundreds of thousands and then rounded. Checking other years after 1975 at random yields correct totals, making it appear as if annual payroll in 1975 truly had a rounding-units miscommunication.

compensation space is that I need to assume all payroll is covered by the program, which is largely true except for various small firm exemptions around the US. I cannot explicitly say how much of the CBP payroll is “covered” by workers’ compensation, though I can cite that more than 97% of payroll and employment was covered by workers’ compensation in 2015 (BLS, 2015; SSA, 2017). This number is generally very stable, with the most recent estimate suggesting 97.5% of all non-federal jobs reported via unemployment insurance filings were covered in 2021 (Welch et al., 2024). Note that there is no one way to measure “covered payroll,” so any estimate is – at best – a rough one.

4 Empirical Methodology

The methodology for this paper exploits a stacked event study design, employing difference-in-difference techniques, following work by Deshpande and Li (2019), Cengiz et al. (2019), and Butters et al. (2022). This setup is beneficial, because it allows for pre-trend testing and the ability to study both contemporaneous effects, and how they develop over time. This is especially true when treatment time varies and concerns mount for the traditional estimator (Goodman-Bacon, 2021); the stacked event study framework avoids problematic (“forbidden”) comparisons that come from allowing treatment time to vary without ensuring clean controls (that themselves do not move in and out of treatment). In the case of competition with a public option in insurance markets, the persistence of concentration effects, specifically, is critical to the discussion of the public option’s long-run behavior. There is also reason to expect heterogeneous effects over time, given each state’s experience before the public option is unique (with each state having its own insurance market).

The general framework for this type of setting is as follows:

$$Y_{s,t,k} = \alpha + \sum_{r=-4, r \neq -1}^{12} \beta_r (PublicOption_s * 1(RelTime_{t,k} = r)) + \gamma_{s,k} + \lambda_{t,k} + \epsilon_{s,t,k}, \quad (1)$$

where the outcome, at least in the main specification, is the workers’ compensation insurance market outcome of interest in state s , year t , and stack k . The event study framework

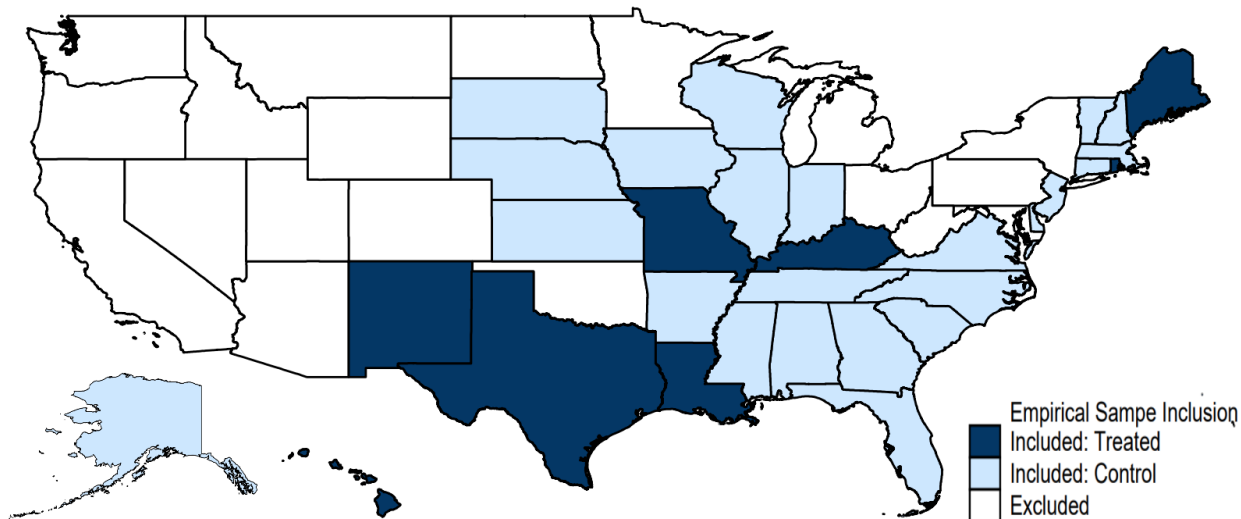
appears with the β_r terms, where we can look back as far as four years prior to a change (with the first one in 1992 and data back to 1988, which is when the NAIC data becomes more consistent) and as far forward as twelve years after a change (given Maine created a public option in 1993 but then made changes to it in 2007, with a buffer for anticipatory effects). If I exclude Maine, the post-period can be expanded to twenty years or more (given Hawaii made the final change in 1997 and data extends through 2020). The *PublicOption_s* variable indicates whether a state is a treatment (*D*) or control (*C*) observation as dictated by whether or not it ever created a public option over the period, which is interacted with the *RelTime_{t,k}* indicators. These relative time indicators track the years before and after public option creation (at $r = 0$), where t tracks the year and k defines the stack's treatment year. The interpretation of the estimates represents the difference between treatment and control states over time, relative to the year just before public option creation.

In the stacked event study framework, only states that are treated (for the first time) during the sample period and states that are never treated during study period are included in the main analysis sample. This means that states changing their market structures in other ways over time are excluded (i.e., Arizona, Michigan, Minnesota, Nevada, and West Virginia). Naturally, this brings up the question as to which market structure is the comparison group of interest: consistently public option states or consistently competitive states? Given the eight treated states in the 1990s that I focus on here all changed from competitive to public option structures, it seems natural to study the public option as a response to a disturbance to the competitive equilibrium. This logic suggests the consistently competitive states are the ideal control group, given they are never treated and align best with the idea of the control group's counterfactual role.

After the restriction away from states changing in ways other than the public option (5) and states that have always had a public option (14), the resulting analytic sample consists of 32 states (with 8 treated states and 24 control states). Figure 3 highlights those states, with white states excluded and blue states included (light blue are consistently competitive

states and dark blue are the states that added a public option between 1992 and 1997).

Figure 3: Empirical Sample Inclusion



Notes: This figure displays state-level inclusion in the main empirical sample, as described in the text. States with no color are excluded; states with color are included (light blue states are control states; dark blue states are treatment states).

In Equation 1 above, there are four stacks (1992, 1993, 1995, and 1997) that include any states creating public options in that year being compared to all consistently competitive states. This provides a set of clean controls that are not affected across the panel, where the only thing changing between stacks is the treatment state(s) included. Ultimately, stacks are mini experiments and the stacked event study specification works across all of them to identify effects in relative time. The specification includes both state-stack and year-stack fixed effects. Clustering is done at the state-by-stack level (implying more than 100 clusters), which is especially important given repeat observations are necessary among never-treated units.

Recent advancements in staggered timing econometrics suggests that not weighting stacked event study estimates with varying treatment group sizes across stacks means the target parameter (i.e., the average treatment effect on the treated or ATT) is not identified (Wing et al., 2024). To identify what the authors call a “trimmed aggregate ATT,”

one needs to apply a weight that corrects for potentially erroneous default weighting of treatment-control trend comparisons. Borrowing from the notation of Wing et al. (2024) and adjusting to match my specification, the weights provided to treatment observations are 1 and to control observations as follows:

$$\frac{N_k^D / N^D}{N_k^C / N^C}, \quad (2)$$

where the $\frac{N_k^D}{N^D}$ term represents the number of states that adopt a public option D in stack k as part of the total number of states ever adding a public option across the study period, N^D . The N_k^C / N^C represents the number of control (never-treated, consistently competitive) states C in stack k relative to the total number of control states across the study period, N^C . This control term simplifies to one because the group of control states is the same in each stack and exhaustive of all possible control units in the study period. This setup produces the parameter of interest:

$$\sum_{k \in \Omega_K} ATT(k, k + r) \frac{N_k^D}{N^D}, \quad (3)$$

where Ω_K represents the trimmed set of public option creation events in the K frame, which consists of four k stack terms (1992, 1993, 1995, and 1997). r , like above, measures event time. Intuitively, the weighting term implies each stack is to be weighted by the fraction of treated units experiencing treatment at that time. Thus, the 1992 stack receives more weight, with half of all treatment states creating their public option at that time. This framework allows this analysis to provide estimates of the average effect of creating a public option in public option states, a useful policy parameter as states modify the roles of their public options.

5 Results

Given the empirical design discussed in the prior section, this section presents a descriptive study of trends, which lends insight to the environment within certain states just before

creating the public option, for each outcome. The outcomes themselves are discussed and motivated in this section, which allows an easier connection between common insurance economics measures with the graphical depictions. Then, I present the formal stacked event study results using the specification from Equation 1 with proper weights.

5.1 Descriptive Analysis

5.1.1 Market Share and Competition

The first outcome of interest focuses on the level of insurer competition in each state marketplace. To measure competition, I use the Herfindahl-Hirschman Index or HHI. This measure is created as follows:

$$HHI_{s,t} = \sum_{f,t} \left(\frac{DPW_{f,s,t}}{StateTotalDPW_{s,t}} \right)^2, \quad (4)$$

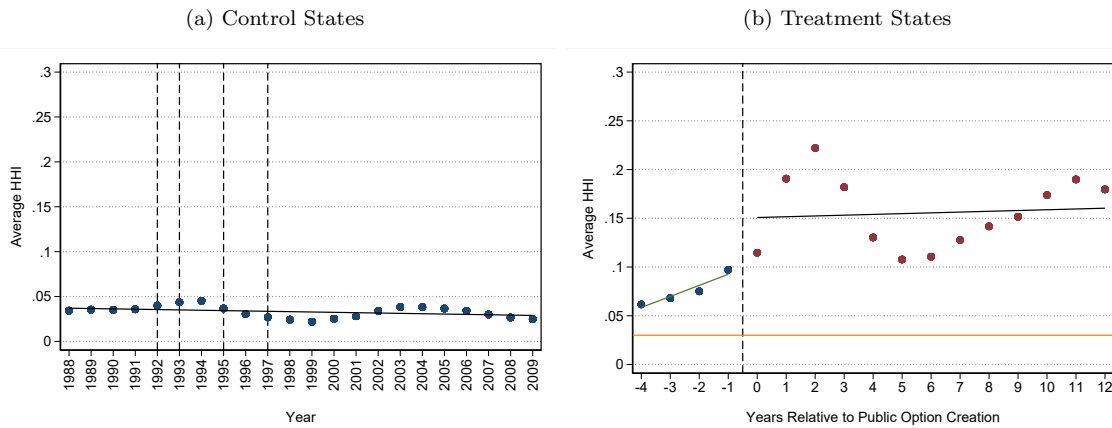
where the HHI in a given state s and year t is provided by placing a firm's (f) workers' compensation direct premiums written (DPW) in a given state and year over the same state's total DPW (across all insurers) in that year. That market share is then squared and summed across all workers' compensation insurers to arrive at state-year HHI observations. This setup excludes insurers who report a negative DPW number, though the results are robust to including those firms.¹¹ A higher HHI value means that more employers in the state are getting their insurance from fewer insurers, thereby increasing the market share of one insurer or a smaller subset of all insurers.

Figure 4 presents trends in HHI among control and treatment states. From panel (a), it is clear that concentration is relatively stable across calendar-year time in control states. Given these are consistently competitive states, this may not come as a surprise. Note that panel (a) runs from 1988 (four years before the first treatment, which was in 1992) through

¹¹Appendix Table A.1 shows that less than 6% of insurer-year observations report negative DPW in workers' compensation. This table also compares negative versus non-negative firm-year observations. Appendix Table A.2 places firms into ever negative and never negative groups to provide comparisons there. As is clear from those comparisons, negative DPW firm-year observations tend to come from smaller stock insurers.

2009 (twelve years after the last treatment, which was in 1997). This is intended to roughly mimic the event time in panel (b) for treatment states, given there is no “event” for control states. For clarity, the four public option creation years are marked with vertical, dashed lines. In control states, the average HHI never exceeds 0.05 and the average across all years (1988-2020) is just under 0.03.

Figure 4: Trends in Concentration (HHI)



Notes: This figure displays average HHIs across states in control states in panel (a) and treatment states (b), across calendar time (year) and event time, respectively. A linear trend line is drawn through all points in panel (a) and drawn separately in panel (b) before and after treatment (at time = 0). Vertical lines in panel (a) indicate public option creation years in treated states. The orange line in panel (b) is the average HHI in control states over all years. Sources: NAIC 1988-2020.

The picture in panel (b), however, is quite different. Graphed against the solid orange line, which is the average HHI in control states across all years, it is not hard to see that treatment states experience an increasingly concentrated marketplace. In fact, the average HHI more than doubles across the pre-period. When the public option joins the market at time 0, there is no noticeable shock to average HHI. Yet, the public options seem to have dramatically increased concentration levels, on average, in the first three years after creation. In the second year after creation, for example, the average HHI in treatment states is nearly five times higher than at the beginning of the study period (and nearly five times higher than in their competitive counterparts). Average HHI declines in subsequent years, but only to the point of elevated concentration from immediately before treatment. In the long run,

it appears new public option states only become increasingly concentrated.

Concerns with the different trends between treatment and control groups are important. There are several key ideas, however, that support the need to assess the response of the market to a public option joining. First, if the goal of governmental intervention in the workers' compensation insurance marketplace was to solve market failure issues associated with increasing concentration, the public option should disturb the upward trend. This means that the ideal counterfactual choice in theory is the competitive market structure, even if there are differences before intervention, because it best represents the idea of where a market is trying to return. Thus, even if readers are skeptical that a positive effect on HHI is causal, a clear rejection of a negative effect is important for policymakers, especially in lines of insurance - like health (GAO, 2022) - that are experiencing increased concentration levels at the same time as debating public options as a solution. Second, continuing with the analysis allows for a more complete discussion on the role of the public option in setting prices, affecting underwriting profitability, and determining market shares.

5.1.2 Underwriting Profitability: Loss Ratios

Situated against this backdrop of continued, elevated concentration - relative to the competitive comparison - in new public option states over time, it is natural to ask about the success of insurers in the marketplace. There are several ways to measure the profitability or performance of insurance companies, given there are premium, expense, and investment avenues that are important. Loss ratios are one of the few measures of performance that can be built back to 1988 using the NAIC data, and they provide insight as to the underwriting performance of an insurer (i.e., the success of the insurer at predicting losses, charging premiums, and paying claims). The state-year average loss ratios are generally depicted as:

$$LossRatio_{s,t} = \frac{DLI_{s,t}}{DPE_{s,t}}, \quad (5)$$

where DLI refers to the average direct losses incurred in a given state among its workers'

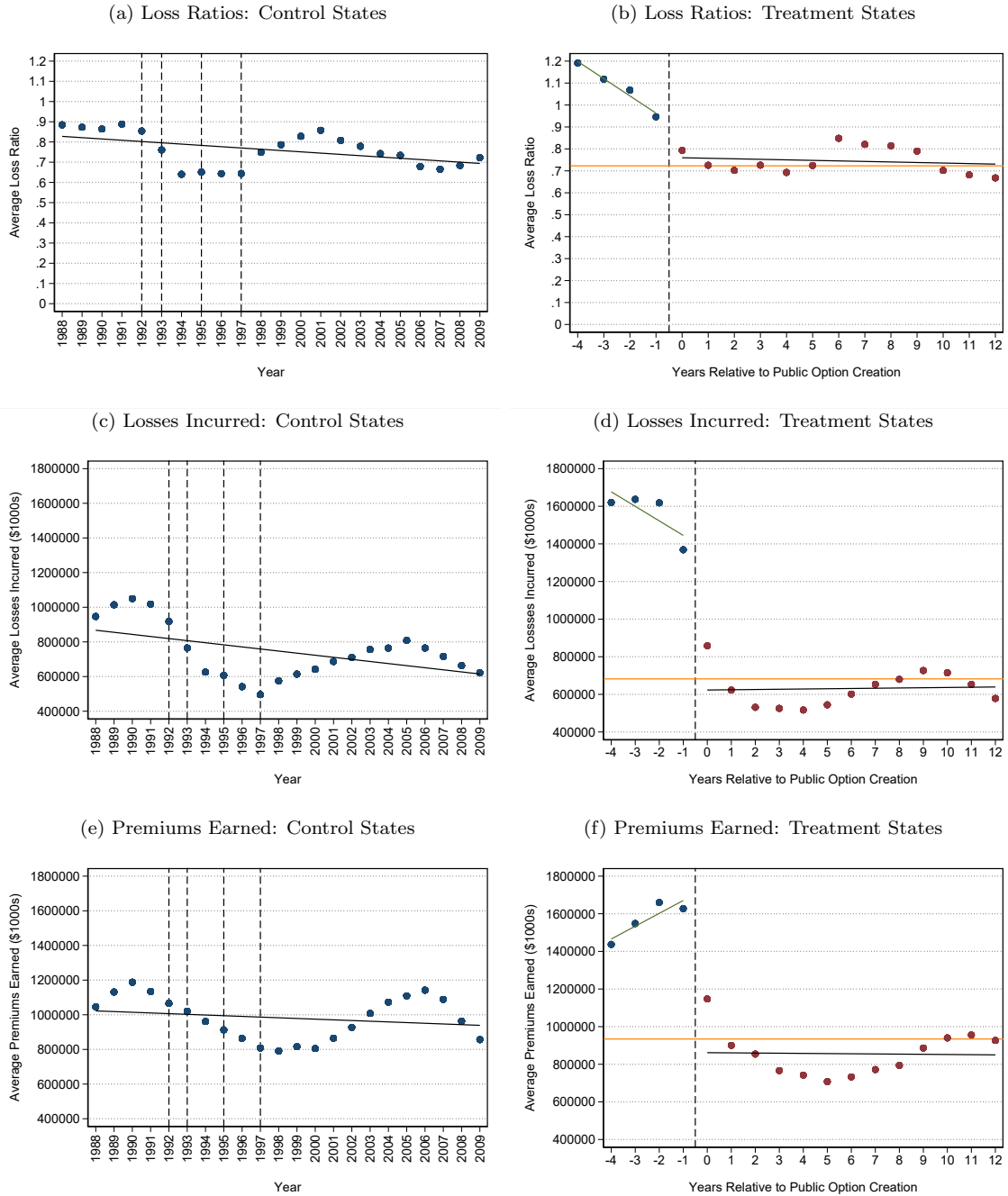
compensation insurers who report positive direct premiums earned (DPE). Thus, this average loss ratio is the ratio of two aggregated values at the state-year level. It is a measure of performance in the sense that, from an insurer’s perspective, a lower loss ratio implies greater profitability. Of course, this is an imperfect measure: it does not account for expenses nor investment returns. Better measures may be the combined ratio (which adds expenses to the numerator in Equation 5) or the operating ratio (which adjusts the combined ratio for an insurer’s investment gains), but the loss ratio does provide basic insights. For instance, a loss ratio over one is not sustainable because, even without expenses included, the insurer is exhibiting issues matching their prices and quantities sold (premiums) to losses. This is especially true at the state level.

The loss ratio results in Figure 5 are situated similarly to the prior figure. I present loss ratios first in panels (a) and (b), before showing the numerator in panels (c) and (d), and then the denominator in panels (e) and (f). Generally speaking, the patterns across all of these graphs are consistent, which is expected for these measures in a period of national consensus on working to reduce statutory benefit levels. Loss ratios are decreasing in similar periods for both treatment and control states, for instance, which tends to be driven by having larger, faster decreases in losses paid (or expected to be paid) than premiums earned. Loss ratios tend to be nearly consistently above 0.9 in pre-period years in both treatment and control groups, with the states eventually adopting a public option having worse experience overall. This unsustainable market eventually recovers. In fact, in all of the treatment state depictions, the period after public option creation is generally related to a return to the competitive average, given by the orange, horizontal line.

5.2 Stacked Event Study Results

Another way to look at these relationships is by using the event study estimates. Figure 6 depicts the resulting point estimates and their 90% confidence intervals after estimating Equation 1. Panel (a) presents the HHI results while panel (b) does the same for loss ratios. In all cases, the reference period is the year before public option creation, which implies that

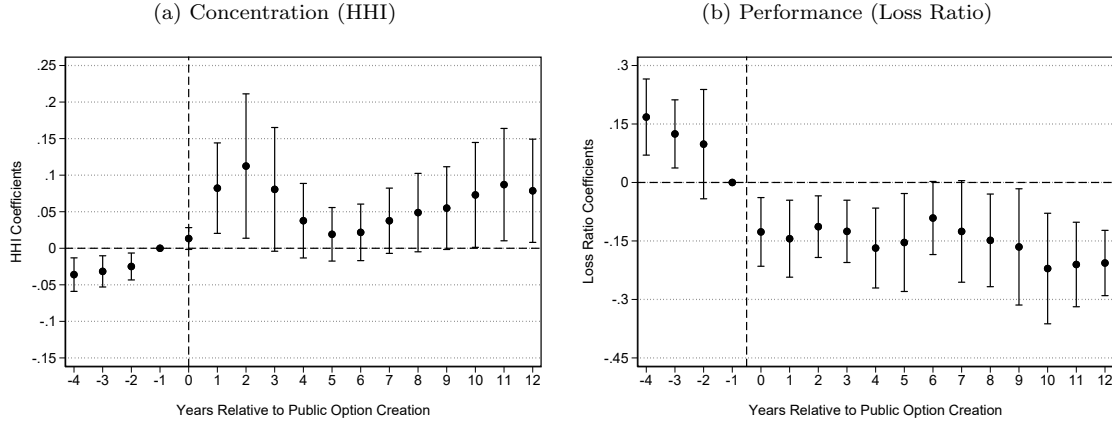
Figure 5: Trends in Underwriting Profitability (Loss Ratio)



Notes: This figure displays average loss ratios, losses incurred, and premiums earned in control states - in panels (a), (c), and (e) - and treatment states - in panels (b), (d), and (f) - across calendar time (year) and event time, respectively. A linear trend line is drawn through all points in control states and drawn separately in treatment states before and after treatment (at time = 0). The orange line in the right-side graphs is the average loss ratio in control states over all years. Vertical lines in the left-side graphs depict public option creation years. Sources: NAIC 1988-2020.

we may interpret these estimates in other years as the gap between the treatment-control difference in any given year relative to how different they were just before public option creation (event time = -1).

Figure 6: Stacked Event Study Results



Notes: This figure displays point estimates from estimating Equation 1 with two outcomes: HHI in panel (a) and loss ratios in panel (b). 90% confidence intervals are used. Note that vertical axes differ based on the difference in outcome variable measured. Weights provided as in Equation 2. Sources: NAIC 1988-2020.

Panel (a) vindicates the earlier descriptive results: differences between treatment and control states in HHI were increasing up to treatment. Particularly notable here is that while there is a trend in coefficients, the pre-treatment confidence intervals do not exclude any other pre-treatment time. So while treatment states are certainly more concentrated in the pre-treatment years, it is only statistically different in the time just before public option creation - and even then the differences are quite small relative to the differences post-creation. The immediate results indicate larger concentration differences between treatment and control states. This effect tends to be persistent with all post-period coefficients being positive. While a few effects are not statistically different from zero in the middle period, all are economically large (given the baseline difference between treatment and control groups was around 0.05 just before the creation of a public option) and all of the confidence intervals suggest that post-period differences are statistically and economically different than pre-period differences.

These results indicate that the public option’s creation did not serve to return a market with diminished competition to one with competition like other states without public options. In fact, the public option seems to have dampened the competitiveness of the market overall, both in the short and long run. The effect was ex-ante ambiguous, where if the goal of the public option was to stoke competition, the coefficients in the post-period should be negative relative to the heightened concentration just before creating a public option. The ultimate effect could have also been expected, especially if one expects the public option to take all of the most risky employers as clients, which mechanically increases its market share but allows private insurers to keep the “good” risks. In the next section, I present evidence to suggest that public options tend to erode market share more than what would be expected if they were just taking “bad” risks, implying that they compete and gain power.

In terms of loss ratios, panel (b) also validates its descriptive trend counterpart above. Here, the level differences in loss ratios are recognized but the trend leading up to the time of a public option being created are not statistically different from one another, and the most immediate coefficient to the comparison period is not statistically different from zero. After public option creation, each year has statistically and economically significant negative effects, implying that - relative to the difference between treatment and control in the period just before creating a public option - new public option states close the gap in underwriting profitability with their competitive counterparts. This result is persistent over the course of time, as well.

There is little reason to have expected underwriting profitability to have worsened in this period, both due to purported competitive influence of the public option and - perhaps more importantly - statutory benefit reductions in most states. The mechanism behind this result, explored in the following section, is important for adding context to the results on concentration but also understanding the public option’s role in affecting insurance prices for employers.

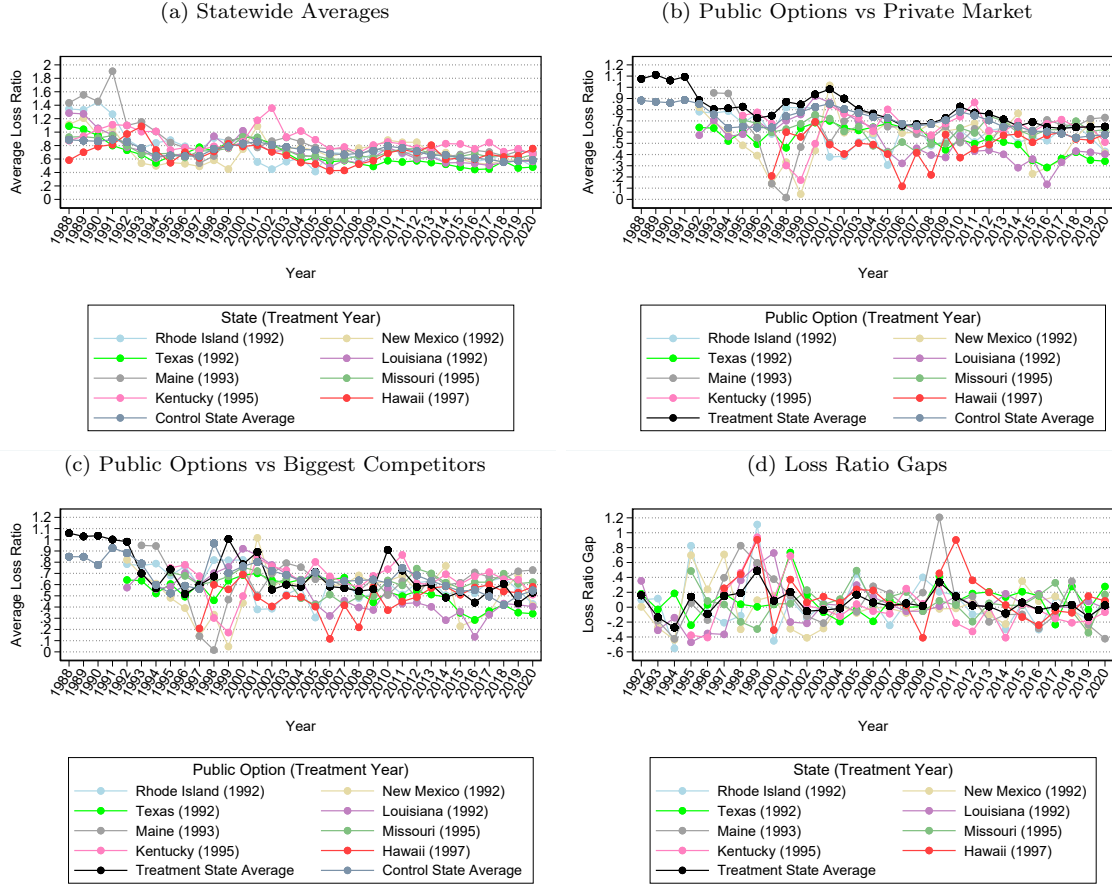
6 Mechanisms & Measuring Price

There are several explanations for the results presented above. The HHI results, for example, could be explained in two ways: a) the public option absorbs “bad risks” as the residual market mechanism (increasing its market share mechanically) and allows the private market to compete in a healthy way or, in a more cynical view, b) the public option erodes competition and steals market share (forcing private insurer exits or acceptance of smaller market shares). The loss ratio results can also be explained in several ways: a) statutory benefit levels were reduced at the same time and were larger reductions than prices, b) shifting of bad risks to the public option (raising its loss ratios) that reduces private insurer loss ratios, c) the public option is able to operate at an advantage that, combining increased market share and higher prices, shifts statewide underwriting profitability with the performance of the public option, and/or d) public option states saw differential shifts in industry mix, particularly away from unsafe industries. One item on this list that cannot be added, however, is that public options join and either deny claims at higher rates or allow the private insurers to do so. In workers’ compensation, there is no way for the insurer to provide less generous benefits than what the state prescribes and the legal foundation for being paid on a workers’ compensation claim as an injured worker remained unchanged. Statutory benefit reductions in this context really mean more controlled healthcare options, less income replacement, and/or a shorter duration of benefits. This section aims to understand these dynamics.

6.1 Ceding “Bad Risks” to the Public Option

One way to understand if the public option is simply accepting the “bad risks” (increasing concentration) while allowing the private market to perform better (lowering statewide average loss ratios) is to compare the loss ratios of the public option to the loss ratios of its private competitors. If this explanation is true, the public option should not be performing better than the rest of the market, given they do not have the ability to reject more claims or decrease benefits for injured workers on their own. Figure 7 explores this dynamic in four panels (using the same colors for each item across the panels).

Figure 7: Loss Ratio Comparisons



Notes: This figure displays loss ratios in panels (a), (b), and (c) over time comparing statewide averages, public option vs. private averages, and public option vs. biggest competitor averages, respectively. Panel (d) graphs a gap between loss ratios of the public option and its biggest competitor. Sources: NAIC 1988-2020.

In panel (a), statewide average loss ratios are graphed separately for each treatment state (including the public option's financial information) and one running control state average. Panel (b) presents the loss ratios of each public option, compared to the private market (i.e., statewide averages after the public option is removed). Panel (c) compares the loss ratios of the public options to their biggest competitors (measured by direct premiums written), using a running average for the treatment group. In this panel, the control state average tracks the loss ratio of the largest insurer in control states. Finally, panel (d) graphs - for each public option state and an overall running average - the loss ratio gap (i.e., $LossRatio_{BiggestCompetitor} - LossRatio_{PublicOption}$).

Panel (a) indicates that the loss ratios in treatment and control states - as seen in the earlier trend graphs - move very similarly after the public option has entered the market, both in trends and levels. Given control state loss ratios are based off a private market that is not forced to sell to “bad risks,” these results indicate that the “bad risks” being included in the public option state averages is not causing deviation from the control state average. Notably, near current time, the workers’ compensation market has been quite successful in its underwriting profitability.

These means mask important variation, though, so panel (b) presents the public option loss ratios separately from the treatment (computed without the public option) and control state average. Here, it appears as though public options are nearly always more successful than the private average loss ratios, not just in the early years but also toward the end of the panel when each public option is more than 20 years old.

Panel (c) does something similar to panel (b) but uses only the biggest insurers, or next biggest if the public option is the largest, to compute the running average. Public options maintain very similar loss ratios to their largest competitors and, while using firm-level loss ratios can be noisy, many years see most public options besting the loss ratio of their biggest competitors. The gap between the public option and its biggest competitor is more clear in panel (d), where the black line is the average gap between the eight treatment states. Despite the noise, the average tracks at 0 or positive in nearly every year, implying that public options are on-track with their biggest competitors that, importantly, don’t need to accept the worst risks.

The evidence from Figure 7 suggests that enhanced underwriting profitability and increased concentration do not come from the public option purely accepting worse risks. Additional evidence supports this idea. Appendix Figure A.1 duplicates the main event study results for loss ratios, with (like Figure 6 and without the public options included). Results indicate that after the first few years, not including the public option nullifies the loss ratio effect, indicating that the public option performance is as good as, and likely better

than, the private market.

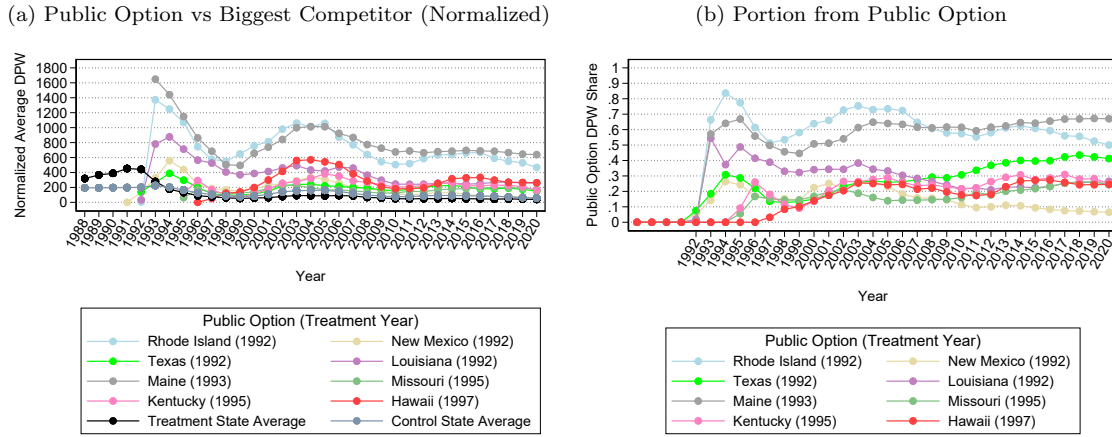
Residual markets are the places where employers that cannot find insurance from the private market (i.e., “bad risks”) receive coverage. As indicated above, most of the eight public options I study here serve as the residual market via guaranteed issue. Data on the size of residual markets is not publicly available, but the National Council on Compensation Insurance (NCCI) operates the residual markets in most states without a public option and has made some residual market summaries available that cover the 1978-1984, 1996-2000, and 2018-2022 periods (NCCI, 1985, 2000, 2022). While these reports do not cover the public option creation period or just before it, residual market shares of all direct premiums written averaged 10.3%, 7.7%, 4.2%, 3.2%, and 3.6% in the 28 states NCCI serviced over the 1996-2000 period, respectively (public option states are not included). In the 2018-2020 period, those values were 7.2%, 6.7%, and 6.2%. Appendix Figure A.2 displays the HHI measure of interest over the entire panel - 1988-2020 - and indicates that the concentration levels of public option states remains high even into the near-present. These HHI measures are inconsistent with simply applying the residual market share from NCCI states to public options (i.e., assuming public options are only writing policies that would be residual market policies).

6.2 Public Option Relevance

Public options are more relevant than just being residual market risk shifting techniques. While the evidence above vindicates that, another way to show this point is to look at the market share, based on direct premiums written (DPW), of the public option. DPW is the key component of HHI as shown in Equation 4. Figure 8 displays market share in two ways: in panel (a), it is the amount of DPW per \$100,000 of statewide annual payroll written by the public option in each state relative to the largest private insurer (depicted as a running average in treatment and control states); and in panel (b), simply the portion of statewide DPW that comes from the public option.

Building off of the discussion in the prior section, these figures provide more evidence

Figure 8: Public Option Market Share - Direct Premiums Written



Notes: This figure displays direct premiums written normalized by statewide payroll in \$100,000s in panel (a). Panel (b) presents the portion of each state’s total direct premiums written that come from the public option. Sources: NAIC 1988-2020; Census CBP 1988-2020.

that the public option is doing more than insuring residual market risks and that the public option’s performance must be a key driver of statewide underwriting profitability. Panel (a) indicates that public options generally step in and provide immediate market support, which is driven in part by what was previously classified as “residual market” business (when benefits were high in the pre-period). Panel (b) shows that every public option consistently writes more than 10% of statewide premiums over the entire 33-year sample period. In fact, even through 2020, seven of the public options have more than 20% market share with three writing more than 40% (Maine, Rhode Island, and Texas).

6.3 Changes in Statutory Benefits and Price Dynamics

While there is little empirical support for the idea that loss ratios decrease because of siphoning bad risks to the public option, statutorily-mandated benefits levels were decreased in many states around the same time as public options were created. There is little reason to expect that data on benefits paid should show a reduction through any other channel than via statute. Thus, one potential explanation is that while benefit levels were reduced, prices did not fall by as much, thus expanding margins and increasing underwriting profitability.

Workers’ compensation benefits paid data, as described above, is different than losses,

because benefits paid reflect all benefits paid in a year for injuries that occurred that year but also any lingering injuries from prior years. Losses reflect losses paid that year but also expectations for lingering injuries in future years. Thus, benefits paid are a fairly accurate depiction of legislated benefit levels but not perfect. Appendix Figures A.3 and A.4 mirror other graphics presented and show trends in normalized benefits paid measures over time (normalization means benefits paid per \$1,000 of annual payroll). These graphics indicate large level differences between treatment and control states in the pre-period and reductions in both ex-post, though larger in public option states. These results suggest this could be a strong mechanism for the loss ratio results, depending on price responses.

Price is not straightforward to measure because *premiums*, what one may think of as price, is a function of both price and quantity. And in many lines of insurance, quantity is an unknown component of premiums because coverage is not always uniform and the number of “units sold” (i.e., clients) are often unknown. In financial data, which is commonly used in insurance economics via the NAIC (as I use in this paper), we do not know how many cars, homes, people, business, etc. are covered by the reported premium amounts. Workers’ compensation, however, presents a unique ability to better estimate price because we largely know the quantity: we know how much payroll exists in each state in each year, and workers’ compensation is required in virtually every state (meaning almost all payroll is covered) and its coverage is standardized (within a state, medical benefits and lost income replacement is consistent across employers). One caveat to all payroll being covered is that some payroll (potentially significant amounts, varying by state) are covered by self-insured employers. Yet another advantage of workers’ compensation is the availability of data on how much self-insured employers pay in benefits in each year and state.

Thus, making use of all the available data, I suggest the following price measure:

$$\text{Price}_{s,t} = \frac{DPW_{s,t}}{\text{InsuredPayroll}_{s,t}}, \quad (6)$$

where the price of workers’ compensation insurance in state s in year t is equal to the total

direct premiums written¹² per dollar of insured payroll, which is defined as:

$$\text{InsuredPayroll}_{s,t} = \text{TotalPayroll}_{s,t} * (1 - \frac{\text{SelfInsuredBenefitsPaid}_{s,t}}{\text{TotalBenefitsPaid}_{s,t}}).^{13} \quad (7)$$

Before moving to the traditional event study estimator to compare the price response and the benefits paid response, it is necessary to provide trend analogs in price. Appendix Figure A.5 indicates broad price reductions in both treatment and control states, with a clear drop after the public option enters the market that returns treatment state unit prices to the competitive price counterfactual.¹⁴ Appendix Figure A.6 presents average prices within each public option state over time, along with running averages of the six treatment states and all control states.

Figure 9 presents formal event study results for price and benefits paid, both per \$1 of annual payroll and graphed on the same set of axes.¹⁵ Panel (a) indicates that prices were reduced by about \$0.005 after the public option entered the market with some variation over time. Panel (b) suggests that the price reduction among private insurers was generally larger, though not always much different, than the overall price reduction. This means that public options are not charging lower prices than the private market and, if anything, are charging higher prices. In other words, absent the public option’s price experience, the price reduction may have been larger, though admittedly the noise in the private price estimates is notable. In panel (c), though, the reduction in benefits paid ranges from \$0.005 in the first year post-creation to between \$0.0075 and \$0.01 for 1-9 years post-creation before moving beyond the \$0.01 mark later in the panel. When taken together, these results indicate that

¹²All elements of this equation are first moved to be in 2020 US dollars using the consumer price index.

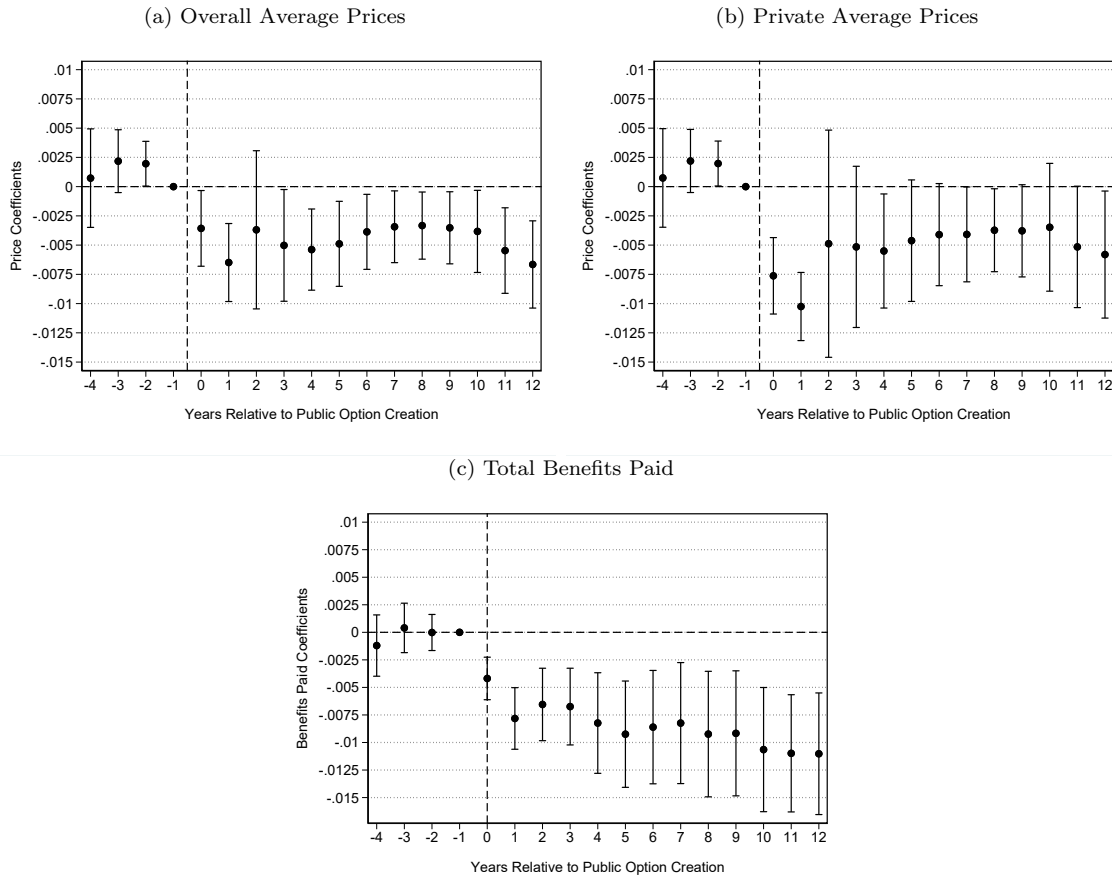
¹³It is possible to modify this setup in several ways. For instance, one could specify insured payroll as privately-insured payroll because insured benefits paid are split by privately-insured and those insured by the public option. One could also adjust the direct premiums written numerator by subtracting dividends paid from the insurer back to the employer, creating an adjusted price measure.

¹⁴It is important to note that these price analyses drop Rhode Island and New Mexico because the benefits paid data has their public options “turning on” in 1996 (four years late), and it is unclear how benefits paid are attributed by payer type between 1992-1995.

¹⁵Appendix Figure A.7 presents event study results across the entire market (like panel (a) of Figure 9) for the self-insured percentage of benefits paid, an adjusted price measure (factoring dividends out of direct premiums written), and dividends paid per \$1,000 of annual payroll.

the loss ratio results may be driven by larger reductions in benefits paid to injured workers than reductions in prices paid by employers. This is true in the short and long term.

Figure 9: Stacked Event Study Results - (Private) Price and Benefits Paid



Notes: This figure displays point estimates from estimating Equation 1 with two outcomes: price in panels (a) and (b), and annual benefits paid in panel (c). All outcomes are per \$1 of annual payroll. Panel (b)'s analysis excludes public options from the analysis. 90% confidence intervals are used. Weights provided as in Equation 2. Sources: NAIC, CBP, SSB, and the National Academy of Social Insurance 1988-2020.

6.4 Entries and Exits

One way to tease out what dynamics are behind the increased concentration of the marketplace is to ask about insurer movement in and out of the market. The public option can gain market share by forcing exits or by actively competing, winning, and gaining market share while the private market accepts a smaller role. In Appendix Figure A.8, I consider the impact of the public option on the net entry ratio, which is the $(N_{Entries} - N_{Exits})/N_{Insurers}$. These results, while indicative of perplexing pre-trends indicate that entries increased after

the public option, more than enough to offset the exit experience. While unlikely to be causal, these results suggest that the public option is not gaining market share by forcing exits. One explanation for these results could stem from the fact that workers' compensation insurers may sell other types of insurance to their commercial clients, allowing them to bundle coverage, so that there is a benefit of remaining in the market even if at a reduced presence.

6.5 Changes in Industry Mix

If public option states saw a differential change in their industry mix away from dangerous work, this could also explain sudden loss ratio improvements. Appendix Figure A.9 studies this dynamic and tests for changes in the percentage of a state's payroll coming from six key industry groups: mining, manufacturing, agriculture/fisheries/forestry, wholesale trade, retail trade, and construction. I use only these six industries because the industries are largely consistent across the CBP data series switching from Standard Industrial Classification (SIC) to North American Industry Classification System (NAICS) codes in 1998. These six categories make up between 40-60% of total payroll in a given state.¹⁶ The results here, while noisy in some industries, largely indicate that industry mix is unlikely to explain the results. While there were some small decreases in mining, there were much larger increases in manufacturing and some increases in construction. There was little change in wholesale and retail trade, and in agriculture, fisheries, and forestry.

7 Conclusion

Public options are often discussed as policy tools in US insurance markets, especially in the health context. Despite the attention, however, there exists little empirical evidence of the effect of public options on competition, underwriting profitability, and prices in this setting. In this paper, I provide (to my knowledge) the first empirical evidence as to the effect of a true public option in US insurance markets by utilizing the workers' compensation arena.

In workers' compensation insurance markets, eight states created public options between

¹⁶Appendix Figure A.10 shows the portion of total payroll captured, including the notable drop in 1998 as the NAICS codes are more granular than SIC codes.

1992 and 1997. I leverage these changes to study the development of competition and insurer underwriting performance in new public option states relative to states that have a purely competitive (only private carriers) market. I document that while new public option states experienced rising concentration leading to the policy change, concentration significantly increased after the public option enters the market. At the same time, underwriting profitability was worse in new public option states before treatment but ultimately returned to the competitive comparators' market average after the public option joined the market.

In testing various mechanisms, I provide evidence that increasing concentration is driven not by the guaranteed issue nature of public options but via active competition and increased market share, much more than can be explained by their residual market role. Insurers are not exiting at higher rates, though, and tend to continue entering the concentrated market, even if only to bundle lines of coverage for commercial clients. The underwriting profitability enhancements cannot be explained by the public option taking “bad risks” to make private insurers more efficient, nor by industry mix changes. It can, however, be explained by changes in price that were smaller than changes in statutory benefit levels.

This work has important policy ramifications in workers' compensation insurance debates. Several states have recently changed their public option rules, with Arizona completely privatizing its public option just over a decade ago and Missouri having just passed legislation to do so for its public option by 2025. Yet, this work also lends insight to the health insurance provision debate. Workers' compensation and health insurance share important features (i.e., largely employer-provided, health risks are protected, concentration issues at the forefront of debate), but they do differ in other ways in current form (workers' compensation insurance is mandated, limiting selection problems, and its benefits are prescribed by statute). With no empirical evidence of a public option in the US health insurance setting, this work may imply that while public options could improve insurer performance and reduce prices, they could come at the cost of increased market power.

References

- Abrams, A. (2019). Where the 2020 Democratic Candidates Stand on Medicare for All. *TIME*.
- Amirkhanyan, A. A., H. J. Kim, and K. T. Lambright (2008). Does the Public Sector Outperform the Nonprofit and For-Profit Sectors? Evidence from a National Panel Study on Nursing Home Quality and Access. *Journal of Policy Analysis and Management* 27(2), 326–353.
- Atal, J. P., J. I. Cuesta, F. González, and C. Otero (2024). The Economics of the Public Option: Evidence from Local Pharmaceutical Markets. *American Economic Review* 114(3), 615–644.
- Baggett, C. B., C. R. Cole, G. Crowley, and E. T. Sirmans (2020). Spillover effects of increased health insurance enrollment on workers’ compensation insurance. *Risk Management and Insurance Review* 23, 53–74.
- Baldwin, M. L. and C. F. McLaren (2016). Workers’ Compensation: Benefits, Coverage, and Costs. Technical report, National Academy of Social Insurance, Washington, D.C.
- Bergman, M. A., P. Johansson, S. Lundberg, and G. Spagnolo (2016). Privatization and quality: Evidence from elderly care in Sweden. *Journal of Health Economics* 49, 109–119.
- BLS (2015). Employment and Wages Online Annual Averages, 2015. Technical report, Bureau of Labor Statistics - Quarterly Census of Employment and Wages, Washington, D.C.
- Born, P., A. Cook, T. Sirmans, and C. Yang (2023). Are health insurers in multiple lines of business less profitable? An examination of scope economies in health insurance. *Journal of Risk and Insurance* 90(1), 185–212.
- Butler, R. J. and J. D. Worrall (1986). The Costs of Workers’ Compensation Insurance: Private Versus Public. *Journal of Law and Economics* 29(2), 329–356.
- Butters, R. A., D. W. Sacks, and B. Seo (2022). How Do National Firms Respond to Local Cost Shocks? *American Economic Review* 112(5), 1737–1772.
- Cabral, M., C. Cui, and M. Dworsky (2022). The Demand for Insurance and Rationale for a Mandate: Evidence from Workers’ Compensation Insurance. *American Economic Review* 112(5), 1621–1668.
- Carroll, A. M. (1993). An Empirical Investigation of the Structure and Performance of the Private Workers’ Compensation Market. *Journal of Risk and Insurance* 60(2), 185–207.
- Caskey, J. P. (2010). The Public Option: Government-Provided Off-Street Parking in Downtown Philadelphia. *William and Mary Policy Review* 1(1), 1–24.
- Cebul, R. D., J. B. Rebitzer, L. J. Taylor, and M. E. Votruba (2011). Unhealthy Insurance Markets: Search Frictions and the Cost and Quality of Health Insurance. *American Economic Review* 101(5), 1842–1871.

- Cengiz, D., A. Dube, A. Lindner, and B. Zipperer (2019). The Effect of Minimum Wages on Low-Wage Jobs. *Quarterly Journal of Economics* 134(3), 1405–1454.
- Census (1988). County Business Patterns, 1974-1986 Technical Documentation. Technical report, U.S. Census Bureau, Washington, D.C.
- Clayman, J. (1959). In Defense of State Workmen’s Compensation Funds. *Rocky Mountain Law Review* 31(4), 428–446.
- Craig, S. V. (2022). Competition in Employer Sponsored Health Insurance: Implications for a Public Option. *Working Paper*.
- Curto, V., L. Einav, A. Finkelstein, J. Levin, and J. Bhattacharya (2019). Health Care Spending and Utilization in Public and Private Medicare. *American Economic Journal: Applied Economics* 11(2), 302–332.
- Danzon, P. and S. Harrington (2001). Workers’ Compensation Rate Regulation: How Price Controls Increase Costs. *Journal of Law and Economics* 44(1), 1–36.
- Danzon, P. M. and S. E. Harrington (1998). *Rate Regulation of Workers’ Compensation Insurance: How Price Controls Increases Costs*. Washington, D.C.: American Enterprise Institute.
- Deshpande, M. and Y. Li (2019). Who Is Screened Out? Application Costs and the Targeting of Disability Programs. *American Economic Journal: Economic Policy* 11(4), 213–248.
- Dillender, M. (2015). The effect of health insurance on workers’ compensation filing: Evidence from the affordable care act’s age-based threshold for dependent coverage. *Journal of Health Economics* 43, 204–228.
- Dinerstein, M. and T. D. Smith (2021). Quantifying the Supply Response of Private Schools to Public Policies. *American Economic Review* 111(10), 3376–3417.
- DWC (2020). Biennial Report to the 87th Texas Legislature. Technical report, Division of Workers’ Compensation, Texas Department of Insurance.
- Eckert, F., T. C. Fort, P. K. Schott, and N. J. Yang (2021). Imputing Missing Values in the US Census Bureau’s County Business Patterns. *NBER Working Paper*.
- Eckert, F., K.-l. Lam, A. R. Mian, K. Müller, R. Schwalb, and A. Sufi (2022). The Early County Business Pattern Files: 1946-1974. *NBER Working Paper*.
- Fishback, P. V. and S. E. Kantor (1996). The Durable Experiment: State Insurance of Workers’ Compensation Risk in the Early Twentieth Century. *Journal of Economic History* 56(4), 809–836.
- Fishback, P. V. and S. E. Kantor (1998). The Adoption of Workers’ Compensation in the United States, 1900–1930. *Journal of Law and Economics* 41(2), 305–341.

- GAO (2022). Private health insurance: Markets remained concentrated through 2020, with increases in the individual and small group markets. Technical report, Government Accountability Office, Washington, D.C.
- Goodman-Bacon, A. (2021, December). Difference-in-differences with variation in treatment timing. *Journal of Econometrics* 225, 254–277.
- Greene, M. R. (1976). The Government as an Insurer. *Journal of Risk and Insurance* 43(3), 393.
- Harrington, S. E. and P. M. Danzon (2000). Rate Regulation, Safety Incentives, and Loss Growth in Workers’ Compensation Insurance. *Journal of Business* 73(4), 569–595.
- Hart, O., A. Shleifer, and R. W. Vishny (1997). The Proper Scope of Government: Theory and an Application to Prisons. *Quarterly Journal of Economics* 112(4), 1127–1161.
- Knutsson, D. and B. Tyrefors (2022). The Quality and Efficiency of Public and Private Firms: Evidence from Ambulance Services. *Quarterly Journal of Economics* 137(4), 2213–2262.
- Kumar, W. M. and E. Y. Adashi (2023). The Colorado Option: The Next Staging Ground for Health Equity? *American Journal of Medicine* 136(1), 9–11.
- Levitin, A. J. and S. M. Wachter (2013). The Public Option in Housing Finance. *U.C. Davis Law Review* 46(4), 1111–1174.
- McLaren, C. F. and M. L. Baldwin (2017). Workers’ Compensation: Benefits, Coverage, and Costs. Technical report, National Academy of Social Insurance, Washington, D.C.
- McLaren, C. F., M. L. Baldwin, and L. I. Boden (2018). Workers’ Compensation: Benefits, Coverage, and Costs. Technical report, National Academy of Social Insurance, Washington, D.C.
- Meyer, B. D., W. K. Viscusi, and D. L. Durbin (1995). Workers’ Compensation and Injury Duration: Evidence from a Natural Experiment. *American Economic Review* 85(3), 322–340.
- Miller, D. P. and J. Yeo (2019). The Consequences of a Public Health Insurance Option: Evidence from Medicare Part D. *American Journal of Health Economics* 5(2), 191–226.
- Mont, D., J. F. Burton, Jr., V. Reno, and C. Thompson (2002). Workers’ Compensation: Benefits, Coverage, and Costs, 2000 New Estimates. Technical report, National Academy of Social Insurance, Washington, D.C.
- Murphy, G., J. Patel, E. Weiss, and L. I. Boden (2020). Workers’ Compensation: Benefits, Coverage, and Costs. Technical report, National Academy of Social Insurance, Washington, D.C.
- Murphy, G. T., J. Patel, L. I. Boden, and J. Wolf (2021). Workers’ Compensation: Benefits, Coverage, and Costs. Technical report, National Academy of Social Insurance, Washington, D.C.

- Murphy, G. T. and J. Wolf (2022). Workers' Compensation: Benefits, Costs, and Coverage. Technical report, National Academy of Social Insurance, Washington, D.C.
- NCCI (1985). 1985 management summary: The workers compensation reinsurance pools. Technical report, National Council on Compensation Insurance, Boca Raton, FL.
- NCCI (2000). Residual market management summary. Technical report, National Council on Compensation Insurance, Boca Raton, FL.
- NCCI (2022). Residual market management summary. Technical report, National Council on Compensation Insurance, Boca Raton, FL.
- Nelson, Jr., W. J. (1988). Workers' Compensation: Coverage, Benefits, and Costs, 1985. *Social Security Bulletin* 51(1), 4–9.
- Nelson, Jr., W. J. (1990). Workers' Compensation: Coverage, Benefits, and Costs, 1987. *Social Security Bulletin* 53(4), 2–11.
- Nelson, Jr., W. J. (1992). Workers' Compensation: Coverage, Benefits, and Costs, 1989. *Social Security Bulletin* 55(1), 51–56.
- Nelson, Jr., W. J. (1993). Workers' Compensation: Coverage, Benefits, and Costs, 1990-91. *Social Security Bulletin* 56(3), 68–74.
- Oh, S. S., I. Sen, and A.-M. K. Tenekedjieva (2022). Pricing of Climate Risk Insurance: Regulation and Cross-Subsidies. *Working Paper*.
- Pang, F., A. Yuen, E. Yoshida, J. Kaya, C. Hane, G. Tsuruda, C. Shima, and R. Kunimune (1993). Workers' Compensation Databook 1993. Technical report, State of Hawaii, Department of Labor and Industrial Relations, Honolulu, HI.
- Polyakova, M. (2016). Risk selection and heterogeneous preferences in health insurance markets with a public option. *Journal of Health Economics* 49, 153–168.
- Price, D. N. (1975). Workmen's Compensation Payments and Costs, 1973. *Social Security Bulletin* 38(1), 34–38.
- Price, D. N. (1977). Workers' Compensation: Coverage, Payments, and Costs, 1975. *Social Security Bulletin* 40(1), 31–36.
- Price, D. N. (1979). Workers' Compensation: Coverage, Payments, and Costs, 1977. *Social Security Bulletin* 42(10), 18–22.
- Price, D. N. (1981). Workers' Compensation: Coverage, Benefits, and Costs, 1979. *Social Security Bulletin* 44(9), 9–13.
- Price, D. N. (1986). Workers' Compensation: Coverage, Benefits, and Costs, 1983. *Social Security Bulletin* 49(2), 5–11.

- Schmulowitz, J. (1995). Workers' Compensation: Coverage, Benefits, and Costs, 1992-93. *Social Security Bulletin* 58(2), 51-57.
- Sen, A. P., Y. Singh, M. K. Meiselbach, M. D. Eisenberg, and G. F. Anderson (2022). Participation, Pricing, and Enrollment in a Health Insurance "Public Option": Evidence From Washington State's Cascade Care Program. *Milbank Quarterly* 100(1), 190-217.
- Sengupta, I., M. Baldwin, and V. Reno (2013). Workers' Compensation: Benefits, Coverage, and Costs, 2011. Technical report, National Academy of Social Insurance, Washington, D.C.
- Sengupta, I. and M. L. Baldwin (2015). Workers' Compensation: Benefits, Coverage, and Costs, 2013. Technical report, National Academy of Social Insurance, Washington, D.C.
- Sengupta, I., M. L. Baldwin, and V. Reno (2014). Workers' Compensation: Benefits, Coverage, and Costs, 2012. Technical report, National Academy of Social Insurance, Washington, D.C.
- Sengupta, I., V. Reno, and J. F. Burton, Jr. (2010). Workers' Compensation: Benefits, Coverage, and Costs, 2008. Technical report, National Academy of Social Insurance, Washington, D.C.
- Sengupta, I., V. Reno, and J. F. Burton Jr. (2005). Workers' Compensation: Benefits, Coverage, and Costs, 2003. Technical report, National Academy of Social Insurance, Washington, D.C.
- Shepard, M., K. Baicker, and J. Skinner (2020). Does One Medicare Fit All? The Economics of Uniform Health Insurance Benefits. *Tax Policy and the Economy* 34(1), 1-41.
- Skolnik, A. M. and D. N. Price (1973). Workmen's Compensation Payments and Costs, 1971. *Social Security Bulletin* 36(1), 31-34.
- Smith, R. S. (1990). Mostly on Monday: Is Workers' Compensation Covering Off-the-Job Injuries? In P. S. Borba and D. Appel (Eds.), *Benefits, Costs, and Cycles in Workers' Compensation*, pp. 115-127. Norwell, MA: Kluwer.
- SSA (2017). Workers' Compensation Program Description and Legislative History. Technical report, Social Security Administration - Office of Retirement and Disability Policy, Woodlawn, MD.
- Thomason, T., T. P. Schmidle, and J. F. Burton, Jr. (2001). *Workers' Compensation: Benefits, Costs, and Safety under Alternative Insurance Arrangements*. Kalamazoo, MI: W.E. Upjohn Institute for Employment Research.
- Weinstein, J. (1967). Big Business and the Origins of Workmen's Compensation. *Labor History* 8(2), 156-174.
- Weiss, E., G. Murphy, and L. I. Boden (2019). Workers' Compensation: Benefits, Coverage, and Costs. Technical report, National Academy of Social Insurance, Washington, D.C.

Welch, T. Q., G. T. Murphy, J. Wolf, and M. Manley (2024). Workers' Compensation: Benefits, Coverage, and Costs. Technical report, National Academy of Social Insurance, Washington, D.C.

Wing, C., S. M. Freedman, and A. Hollingsworth (2024). Stacked Difference-in-Differences. *NBER Working Paper*.

Chapter I Appendix Tables and Figures

Intended for online-only publication.

Table A.1: Comparing Non-Negative DPW Firm-Years to Negative DPW Firm-Years

	(1) Non-Negative	(2) Negative
Member of Group	0.92 (0.28)	0.96 (0.20)
Stock Insurer	0.86 (0.35)	0.92 (0.27)
Mutual Insurer	0.11 (0.31)	0.07 (0.25)
Magnitude of Negative (\$1000s)	0.00 (0.00)	-266.99 (1450.54)
Direct Premiums Earned (\$1000s)	4792.06 (23542.25)	-112.99 (1640.76)
Dividends (\$1000s)	143.46 (2712.35)	13.02 (335.86)
Direct Losses Paid (\$1000s)	2659.60 (14003.95)	981.28 (6214.04)
Loss Adjustment Expenses Paid (\$1000s)	268.10 (1564.76)	108.65 (773.28)
Direct Premiums Written (\$1000s)	4848.97 (23711.46)	-266.99 (1450.54)
<i>N</i>	554928	30764

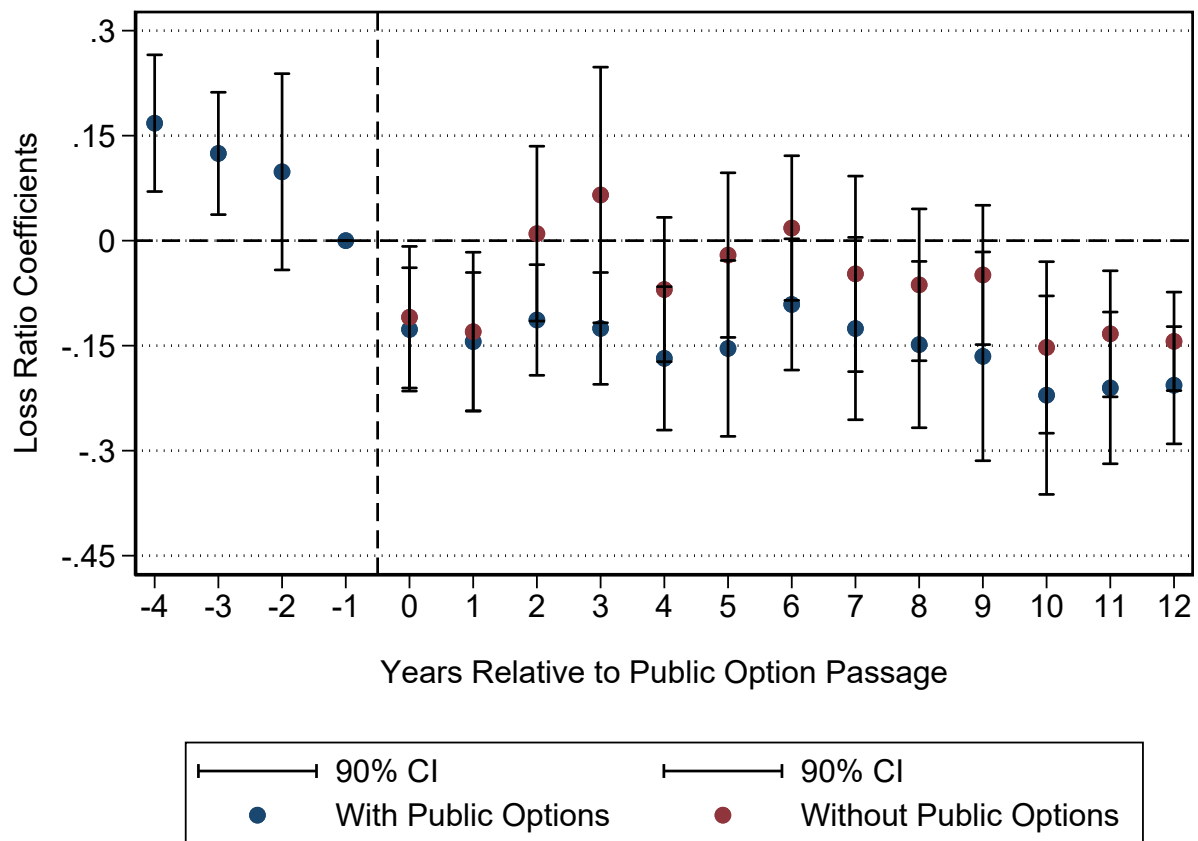
Notes: Depicted are sample averages across variables (rows) and negative DPW status (columns). Standard deviations are in parentheses. Dollar amounts in 2020 US dollars. Source: NAIC 1984-2020.

Table A.2: Comparing (N)ever Negative Firms

	(1) Non-Negative	(2) Negative
Member of Group	0.89 (0.32)	0.96 (0.21)
Stock Insurer	0.82 (0.38)	0.90 (0.30)
Mutual Insurer	0.13 (0.34)	0.08 (0.27)
Direct Premiums Earned (\$1000s)	6675.36 (30669.98)	2634.84 (12812.70)
Dividends (\$1000s)	213.66 (3700.53)	69.63 (1119.49)
Direct Losses Paid (\$1000s)	3612.94 (18586.72)	1668.64 (7260.79)
Loss Adjustment Expenses Paid (\$1000s)	351.40 (1979.47)	179.92 (1001.44)
Direct Premiums Written (\$1000s)	6764.11 (30860.66)	2639.22 (12959.98)
Avg. of Neg. DPW Firm-State-Years when Negative	. (.)	-260.63 (1019.66)
Avg. of Neg. DPW Firm-State-Years when Positive	. (.)	3063.51 (9642.29)
<i>N</i>	315992	269700

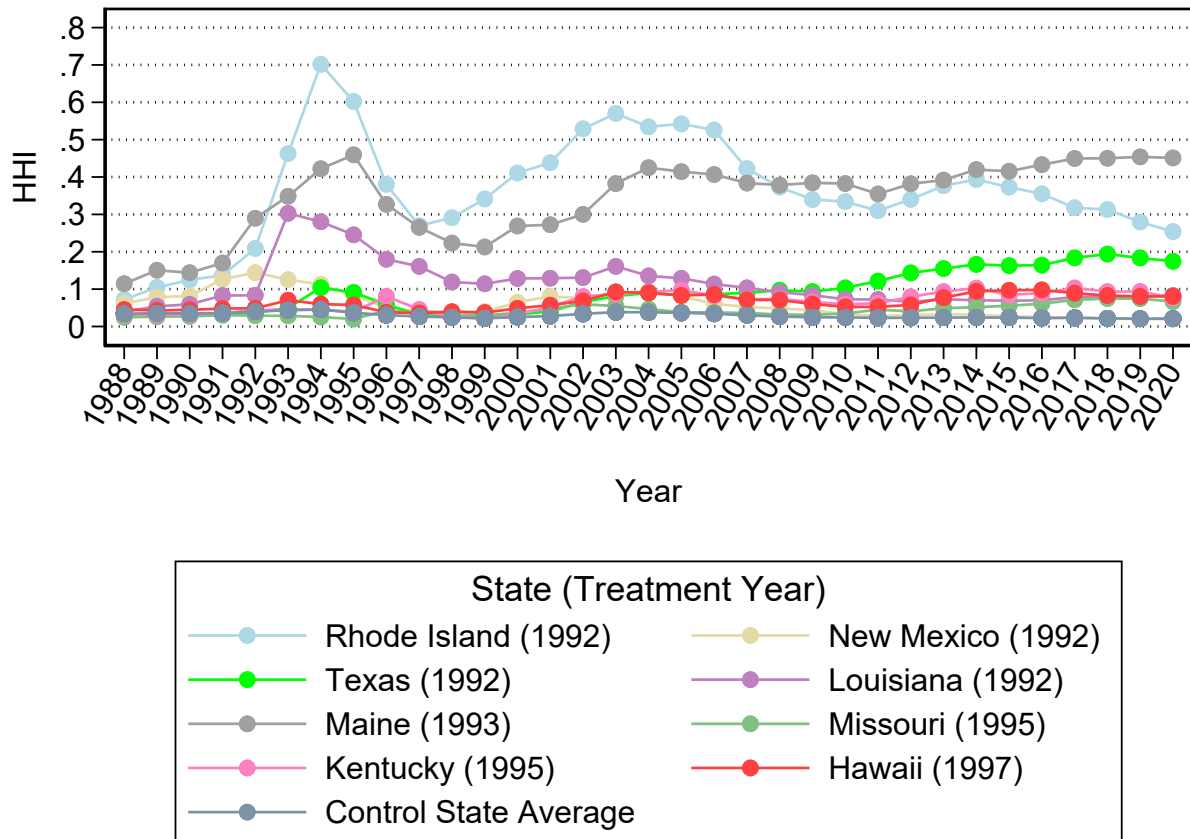
Notes: Depicted are sample averages across variables (rows) and (n)ever negative DPW status (columns). Standard deviations are in parentheses. Dollar amounts in 2020 US dollars. Source: NAIC 1984-2020.

Figure A.1: Loss Ratio Results With(out) Public Options



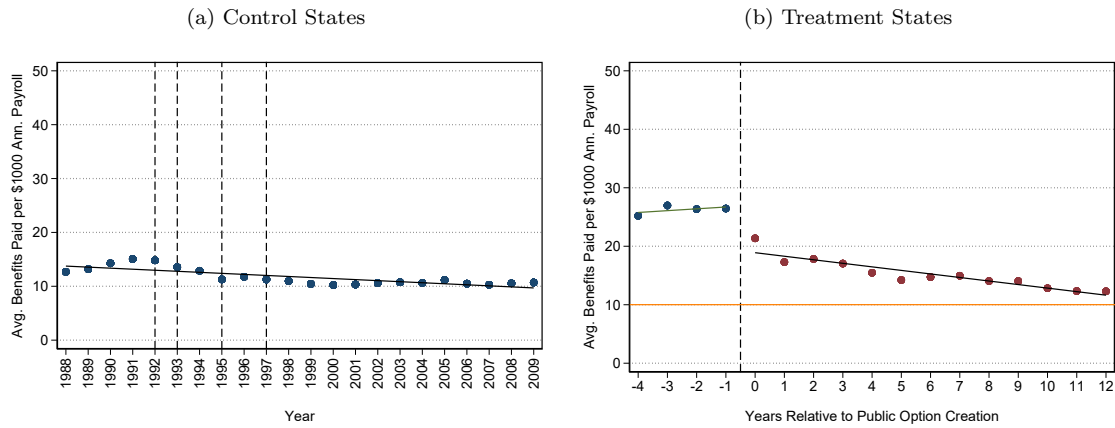
Notes: This figure displays point estimates from estimating Equation 1 with respect to loss ratios. 90% confidence intervals are used. Weights provided as in Equation 2. Blue dots duplicate panel (b) of Figure 6, while red dots repeat that analysis after dropping the public options. Sources: NAIC 1988-2020.

Figure A.2: HHIs Over Time



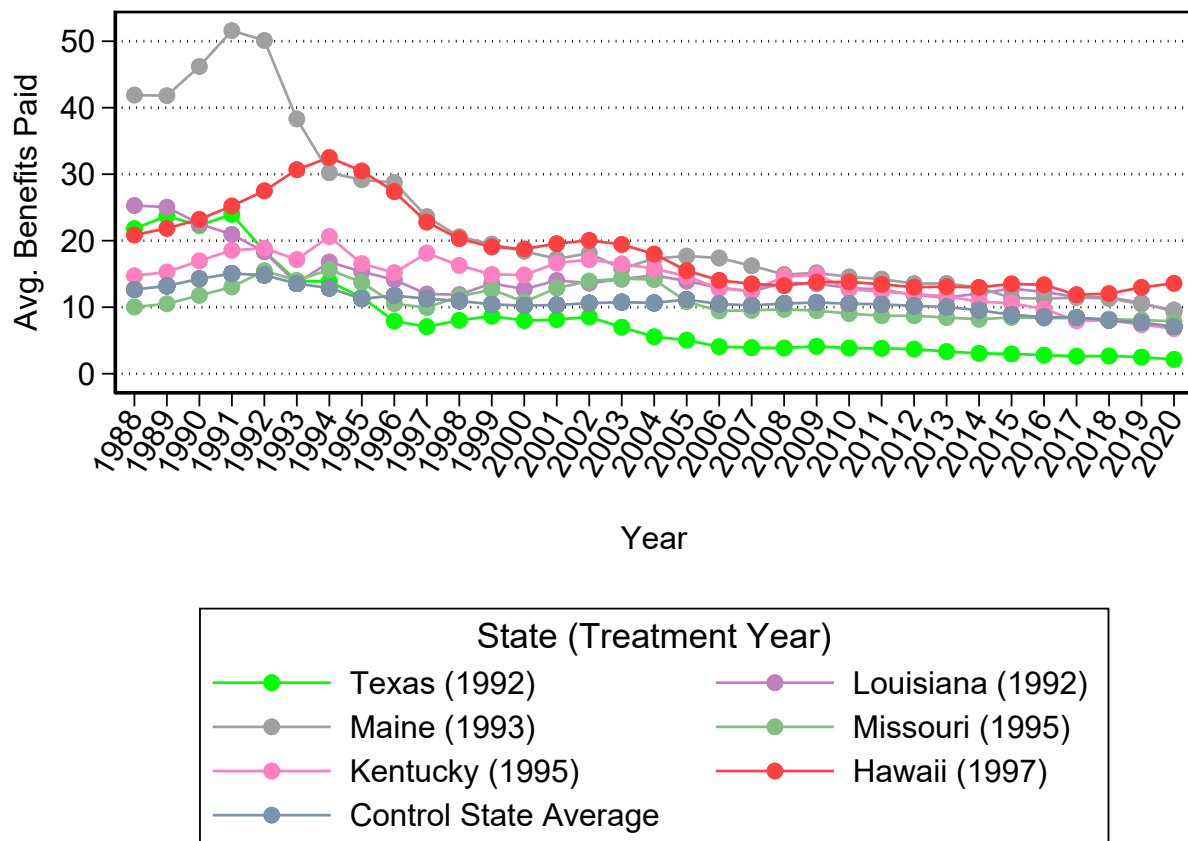
Notes: This figure displays HHI measures in each public option state, and via a running average from control states, across time. Sources: NAIC 1988-2020.

Figure A.3: Trends in Benefits Paid



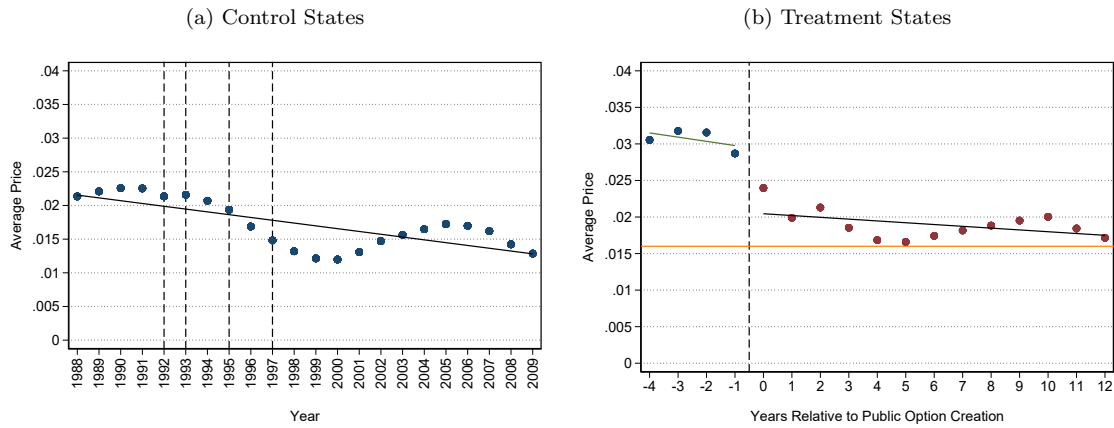
Notes: This figure displays average workers' compensation benefits paid per \$1,000 of annual payroll within control states in panel (a) and treatment states (b), across calendar time (year) and event time, respectively. A linear trend line is drawn through all points in panel (a) and drawn separately in panel (b) before and after treatment (at time = 0). The orange line in panel (b) is the average benefits paid in control states over all years. Vertical lines in panel (a) indicate public option creation years. Sources: CBP, SSB, and the National Academy of Social Insurance 1988-2020.

Figure A.4: Benefits Paid by State



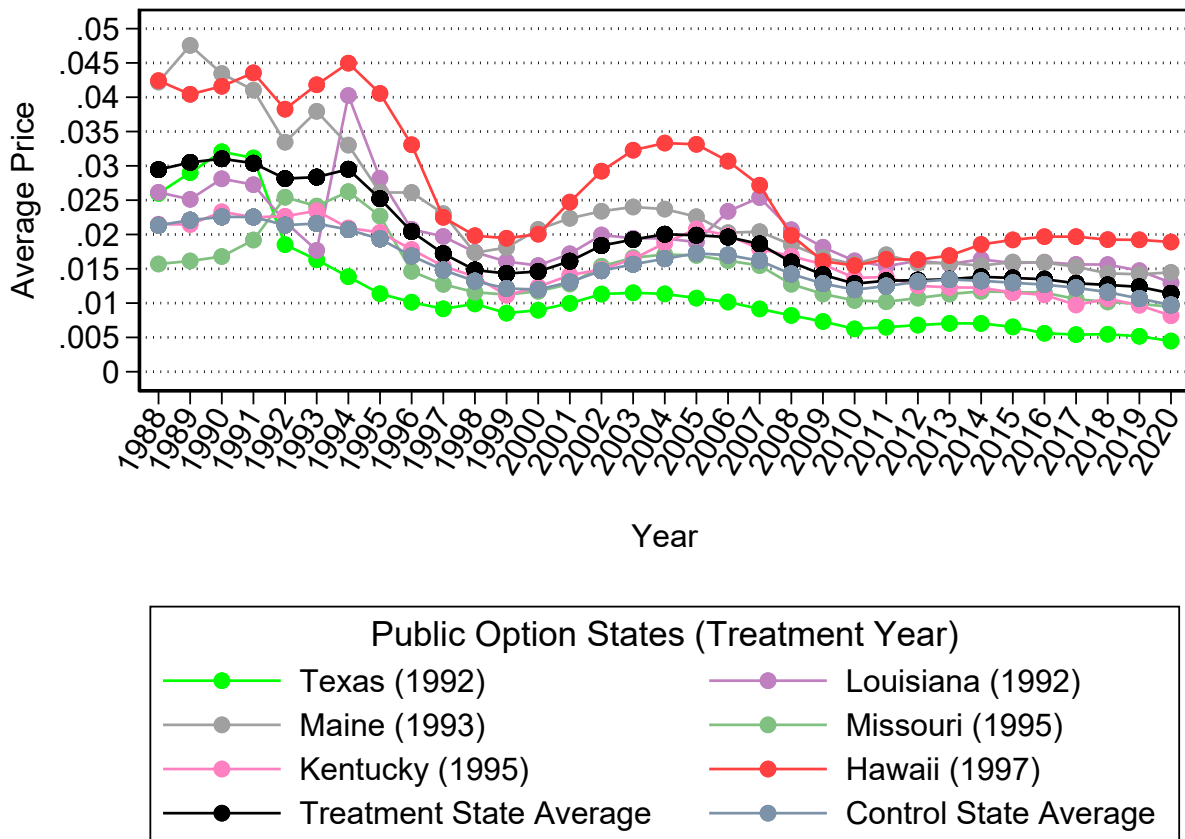
Notes: This figure displays average workers' compensation benefits paid per \$1,000 of annual payroll within each public option state and a rolling average of control states. Sources: CBP, SSB, and the National Academy of Social Insurance 1988-2020.

Figure A.5: Trends in Price



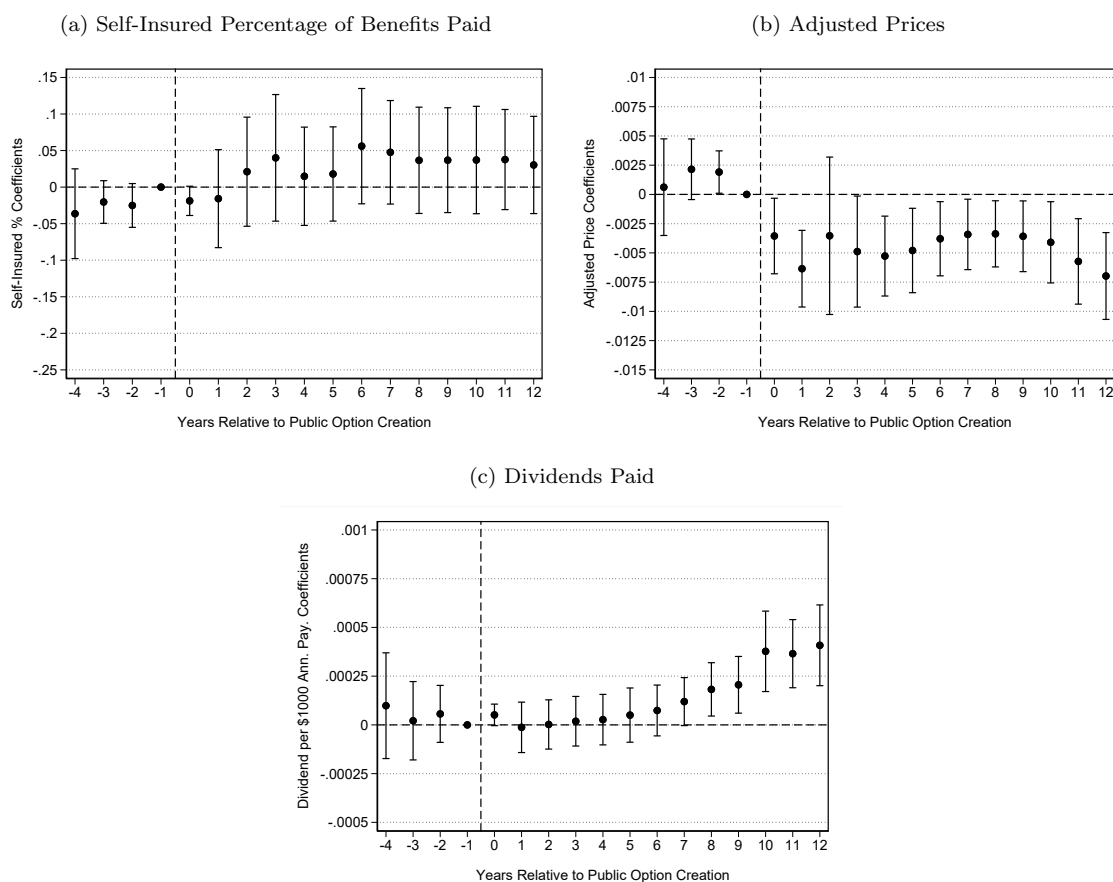
Notes: This figure displays average workers' compensation price (per \$1 of annual payroll) within control states in panel (a) and treatment states (b), across calendar time (year) and event time, respectively. A linear trend line is drawn through all points in panel (a) and drawn separately in panel (b) before and after treatment (at time = 0). The orange line in panel (b) is the average benefits paid in control states over all years. Vertical lines in panel (a) indicate public option creation years. Sources: NAIC, CBP, SSB, and the National Academy of Social Insurance 1988-2020.

Figure A.6: Prices by State, Year



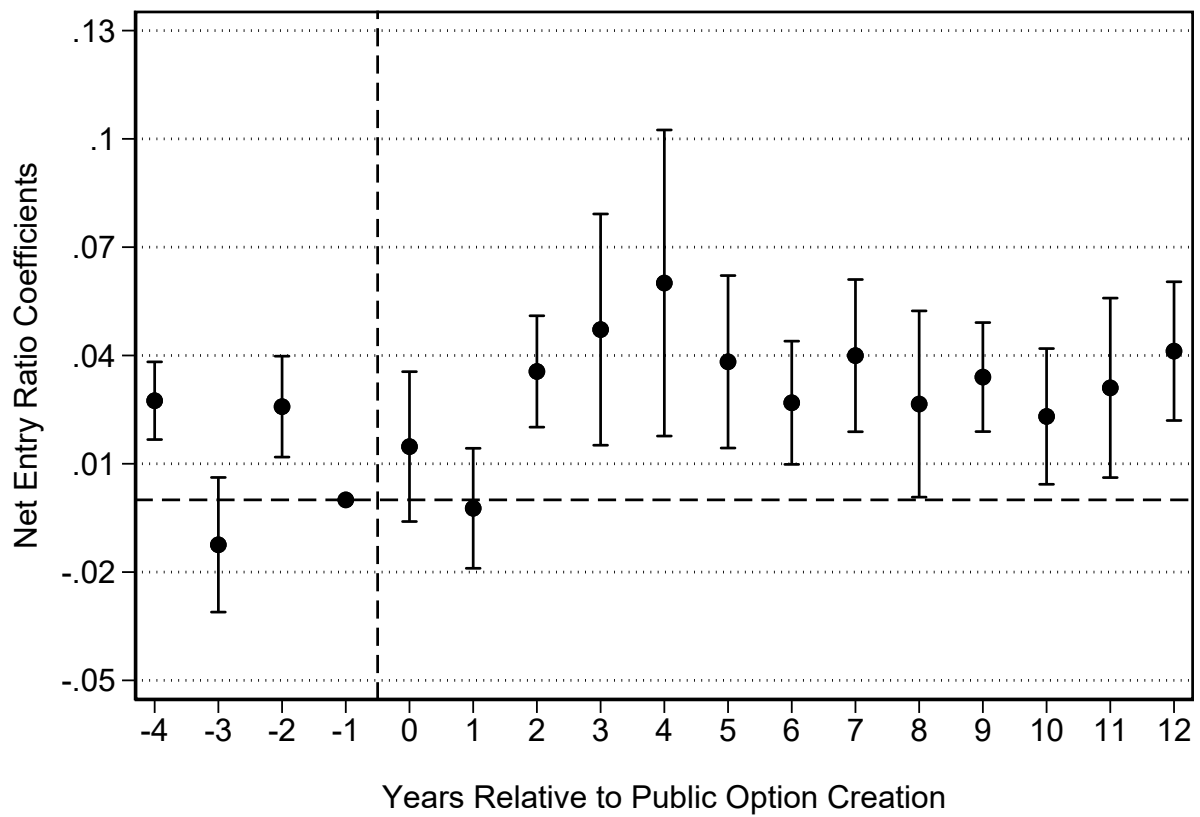
Notes: This figure displays average prices by treatment state, along with a running average for treatment and control states, over time. Sources: NAIC, CBP, SSB, and the National Academy of Social Insurance 1988-2020.

Figure A.7: Stacked Event Study Results - Self-Insured Benefits, Adjusted Price, Dividends



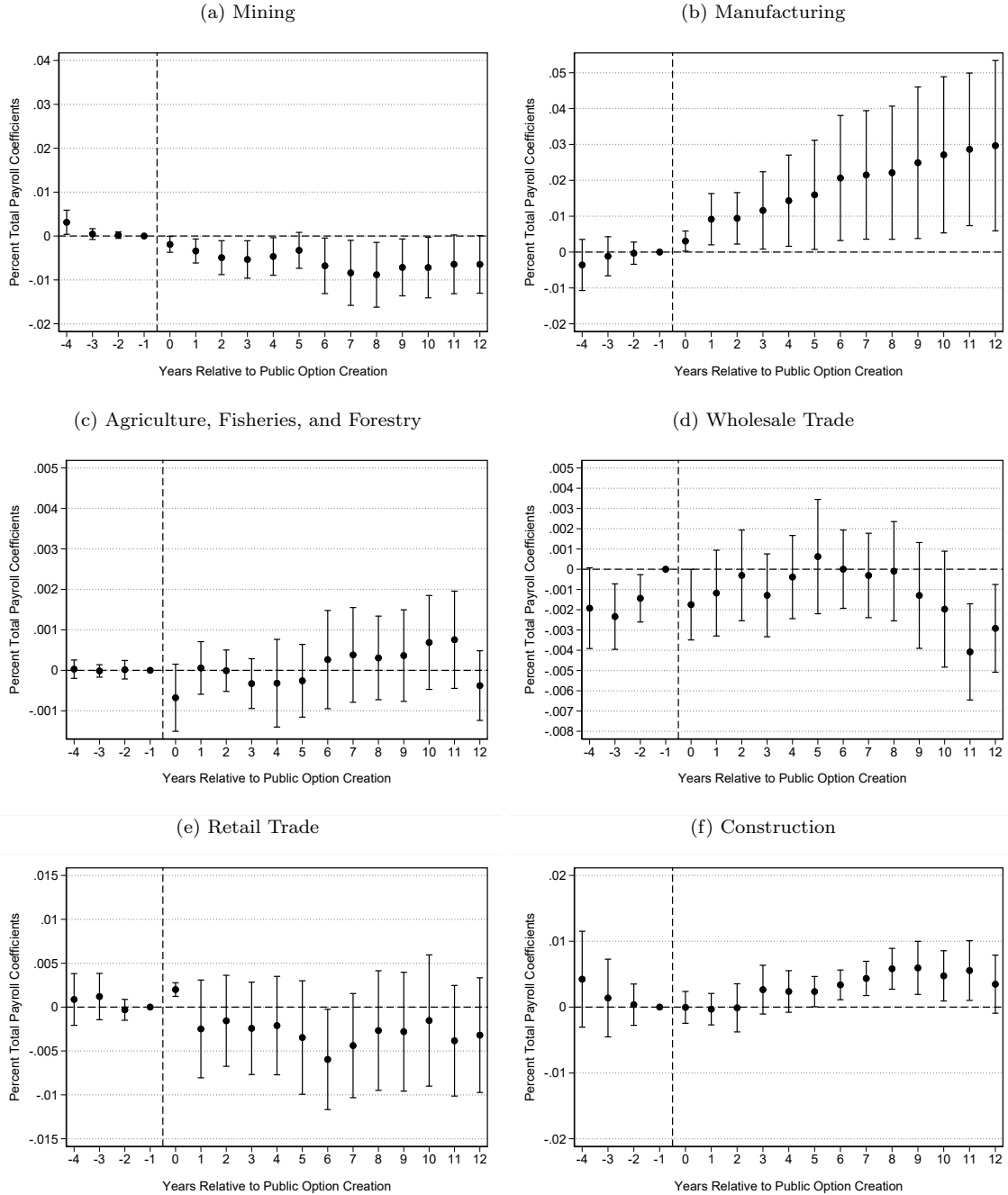
Notes: This figure displays point estimates from estimating Equation 1 with three outcomes: self-insured percentage of total benefits paid in panel (a), prices adjusted for dividends paid in panel (b), and then dividends paid per \$1,000 of annual payroll in panel (c). 90% confidence intervals are used. Weights provided as in Equation 2. Sources: NAIC, CBP, SSB, and the National Academy of Social Insurance 1988-2020.

Figure A.8: Net Entry Ratio



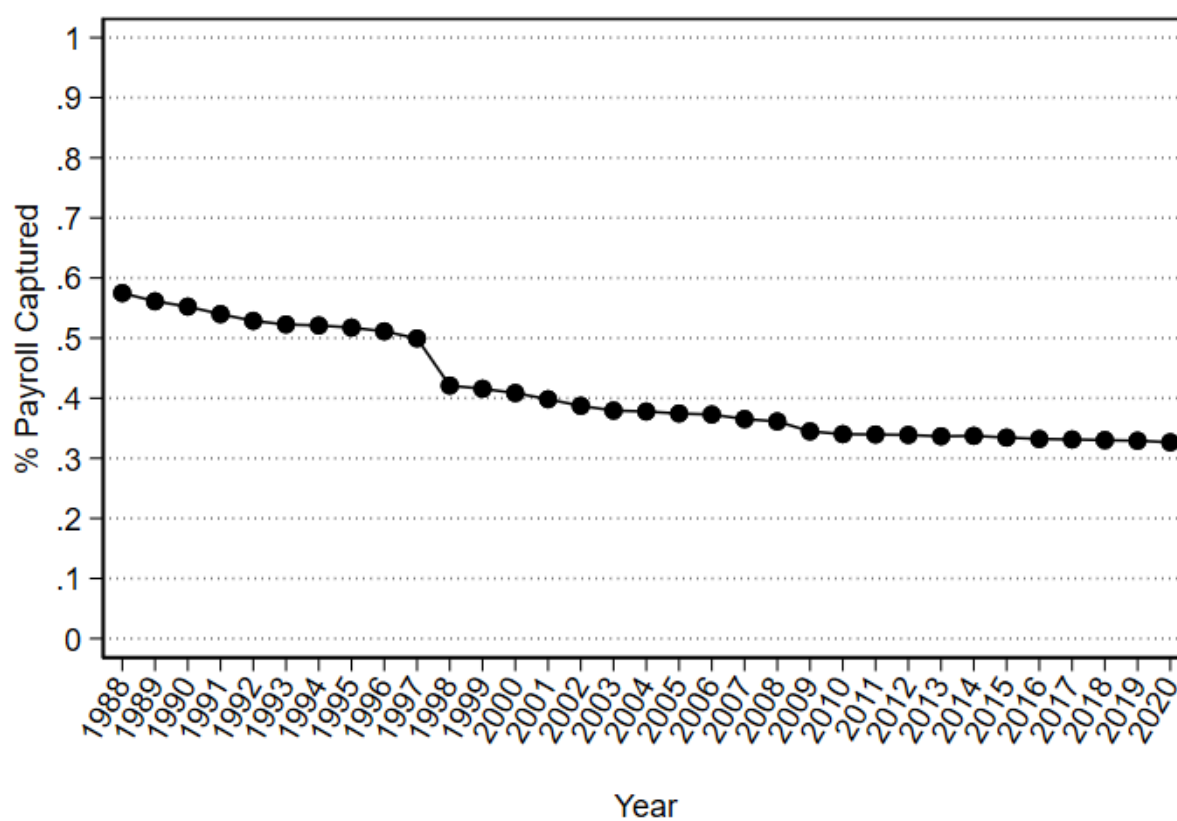
Notes: This figure displays point estimates from estimating Equation 1 with respect to the net entry ratio. 90% confidence intervals are used. Weights provided as in Equation 2. Sources: NAIC 1988-2020.

Figure A.9: Stacked Event Study Results - Industry Mix



Notes: This figure displays point estimates from estimating Equation 1 with the outcome as the percentage of a state's total payroll coming from the industry named in each sub-figure title. 90% confidence intervals are used. Weights provided as in Equation 2. Sources: CBP 1988-2020.

Figure A.10: Payroll Captured



Notes: This figure shows how much of total payroll, on average, is captured by the six broad categories from Appendix Figure A.9, over time. Sources: CBP 1988-2020.

Chapter II -

Choice as Public Policy: Evidence from Optional Workers' Compensation Coverage in Texas*

Abstract

Do workers benefit when workers' compensation insurance is optional for employers to purchase? Using the unique setting of Texas as the one state in the US with a robust opt-out for employers, I study the effect of losing coverage on wages, employment, and injuries. I also provide evidence about the social cost of optional coverage in the form of lawsuits and personal injury firm advertising, given employees can sue their employers for work injuries when an employer is uncovered. To get around the endogeneity-laced firm coverage decision, I propose an instrument for coverage in the form of local, partisan judge elections where a non-incumbent judge or party wins. Preliminary results indicate that wages are positively related to coverage using ordinary least squares, which is in opposition to the traditional theory. Two-stage least squares methods are then discussed. This work is important for recent policy debates where several states have sought to mimic Texas' optional system.

Keywords: workers' compensation, public policy, insurance, coverage mandates

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1 Introduction

While it is well-accepted that workers pay for more generous workers' compensation benefits via lower wages (Gruber and Krueger, 1991), it is less clear what happens when they lose their workers' compensation coverage. The economics literature would, via compensating differentials, predict higher wages for uncovered workers (Rosen, 1974, 1986). Yet, there are reasons to think uncovered workers do not demand higher wages, such as undervaluation of workers' compensation benefits, labor market preferences, lack of salience, or poor understanding of the lawsuit-driven outside option (Auerbach et al., 2010). In addition, uncovered workers could come at an increased social cost as personal injury lawsuits become the main avenue by which injuries are compensated. These dynamics are further intensified by recent work suggesting little rationale for a coverage mandate (Cabral et al., 2022) and efforts by legislators in several states to turn their mandated system into an optional one.¹

To gain insight into this effect of losing coverage, I study the one state in the US with a robust opt-out of workers' compensation insurance for employers: Texas. In other states, it is not possible to study the effect of not being covered by workers' compensation, both because of data access and the fact that only small employers (between one and five employees) may be able to take a coverage exemption. In Texas, however, 25% of employers and 17% of employees were not covered in 2022; 20% of employers with 500+ employees were uncovered (DWC, 2022). In an effort to better understand the consequences of losing coverage, I make use of several unique administrative datasets constructed via open records requests in Texas (similar in nature to Cabral et al. (2022)). The crux of the analysis is built off of legally-required filings for uncovered employers each year, allowing me to identify changes in uncovered rates at the industry-county level over the 2010-2022 period. These changes are then supplemented with publicly-available data to understand relationships with wages, employment, injuries, lawsuits, and personal injury firm advertising.

¹Oklahoma, Tennessee, Florida, Arkansas, and South Carolina have all proposed bills to do so, with Oklahoma even passing one before the state Supreme Court found it unconstitutional. The Department of Labor (DOL) has recently expressed concern with this trend: <https://blog.dol.gov/2022/11/18/3-concerning-trends-for-injured-and-ill-workers>.

In every state, an employer is considered covered when they provide workers' compensation coverage to employees, either via buying insurance or organizing a self-insurance arrangement certified by state regulators (where it is allowed). By being covered, the employer and its employees are "subscribers" to the workers' compensation system defined by two key pillars: 1) employees are given near guaranteed payment for injuries at work in the form of lost income replacement (not full) and medical care; 2) employers are generally protected from employees filing lawsuits for larger damage amounts. Thus, in Texas, the choice to be a subscriber lives with the employer, and when an employer is a non-subscriber, they lose their legal protections and can be sued by their employees (even if they offer alternative injury plans, generally). Given that - in 2022 - only 30% of non-subscribed employers offered alternative injury plans, and 89% used a reporting window of 24 hours or by the next day (subscribers in Texas have a 30-day reporting window), the non-subscriber decision shifts substantial risk to employees ([DWC, 2022](#)). This presents a unique laboratory to understand the effects on workers and society.

This paper seeks to answer several research questions by exploiting the open records request information and this Texas laboratory. The guiding question to which this paper aims to be a partial answer is: do employees win or lose when their employer has the choice to be covered or not? This is a broad question, and I provide several categories of answers, starting with asking about the labor market responses, specifically in terms of employment and wages. These outcomes are also relevant to connecting to questions about how the theory of compensating differentials applies to an oft-overlooked employee benefit.

I then ask about how societal outcomes are driven by the employee experience, namely via lawsuits and personal injury firm advertising. Societally, lawsuits are likely to be an inefficient way to allocate work injury costs, when costs are incurred immediately but remedy may not be offered for many years. An influx of personal injury cases into the court system is also likely to cause negative externalities, by increasing wait times for other cases via judge caseloads, stretching county budgets, and slowing a slow judicial system.

Finally, I ask about the injury experience of newly-covered firms to understand more about the injury experiences along the coverage spectrum. This question is motivated by the fact that, in 2022, 22% of all manufacturing employers are uncovered, along with 16% of mining employers and 29% of those in agriculture; these numbers among riskier work stands out considering only 22% of employers in the finance, real estate, and professional services industries are uncovered (DWC, 2022). This evidence serves as a roadmap for a deeper understanding of the injury experiences for employees at covered and uncovered employers.

Using uncovered filings with the state in the 2010-2022 period, I identify which county-industry observations experience changes in their uncovered rate and present ordinary least squares (OLS) results showing that being in an industry-county pair that experiences an uncovered rate increase is associated with decreases in total wages that are statistically significant - with a coefficient of about \$213,900 - with corresponding statistically significant decreases in average weekly and annual wages. These results imply while employment may have decreased, employment-adjusted wages also decreased. These results are perplexing from the theory perspective because the loss of a non-cash compensation component should increase wages, if individuals understand workers' compensation and it's fully salient.

Yet, it is possible that the firm coverage choice is endogenous for several reasons, including injuries, and that the uncovered rate could be measured with error. To deal with these concerns, I propose a two stage least (2SLS) estimation where I instrument for the uncovered rate with local judge elections, which are partisan in Texas, where the court's judge (either with an incumbent loss or retirement) or, more specifically, party changes. This instrument is plausibly exogenous in the sense that the only reason a new judge should affect wages or employment is by changing an employer's calculus of the tradeoffs of being uncovered. For instance, a judge that tends to be friendlier to workers should push more firms to be covered, given they face more legal liability risk from being uncovered after that time. I do not take a stance on which party or type of judge should be more or less friendly to workers and only make use of the judge-driven shock.

In the first stage, changes to the benches in a district - either with new judges alone or also with new parties - appear as the strongest instruments (with F -statistics near or above 50), compared to other variables like the percent of a county's judges that are Republican and proxies for how competitive a race was. In terms of new judges and new parties, the variables are significantly negative in explaining increases in uncovered rates. This supports the idea that more volatile counties in terms of judges tend to have fewer uncovered employers and serves to isolate the component of being uncovered that should most directly impact wages: changes in the employers' tradeoffs.

This study contributes to the literature in several ways. First, this paper contributes to the body of work focused on the relationship between workers' compensation programs and labor market outcomes, most often wages. Several papers provide important results about how workers' compensation benefits shift wages that are consistent with compensating differentials² (Viscusi and Moore, 1987; Moore and Viscusi, 1990a,b; Kaestner and Carroll, 1997; Gruber and Krueger, 1991; Fishback and Kantor, 1995).³ Gruber and Krueger (1991) present the most renowned work in this area, showing that benefits and wages offset nearly dollar for dollar, and their analysis forms the crux of the argument that Cabral et al. (2022) use to justify their approach, which leads to finding little reason for coverage mandates. My analysis contributes to this literature by being the first, to my knowledge, to study the loss of coverage altogether. Each of the other analyses assesses marginal, continuous changes in benefit levels, but always within states using coverage mandates. The one exception is the seminal work of Fishback and Kantor (1995), wherein they show that initial creation of

²The theory of compensating differentials between non-wage job characteristics and wages has been explored widely (and will not be studied here, in that way), with mixed evidence and some (surprising) relationships (Brown, 1980; Sorkin, 2018; Wissmann, 2022). Brown (1980) reviews early work in this area and chalks up mixed evidence and wrong-signed relationships to bias from omitting personal characteristics. His own analysis ended up refuting that hypothesis. Sorkin (2018) provides a more recent review and discusses the challenges in this literature. Important recent work, though, provides clear evidence of compensating differentials using an identification strategy rare in this setting (Wissmann, 2022). Compensating differentials relating to quit rates in the work context have also been explored (Viscusi, 1979), as have changes in employers' wage costs (Holmlund, 1983).

³For interested readers, Krueger and Meyer (2002) review empirical evidence in workers' compensation including the effects of benefits level changes on injuries, both on the intensive (how long they last) and extensive (incidence rates) margins.

workers' compensation programs led to decreases in wages.⁴ Thus, my work asks about if the relationship works in the opposite (and extreme) direction.

This paper also contributes to the more narrow literature of the operation of workers' compensation markets with optional coverage. The only state with a robust opt-out is Texas, with the last state to make the change to mandatory coverage being South Carolina in 1997 (Morantz, 2010), with all other states mandating coverage at least since 1987 (Nelson, 1990). Jinks et al. (2020) provides evidence that the main reason for an employer to become a non-subscriber in Texas is costs, creatively comparing injury rates within multi-state employers who are making an uncovered choice in Texas that, by definition, must be unique to the firm across its operating states. Butler (1996) also studies the injury experience of non-subscribed employers in Texas. The most relevant paper in this area to mine, however, is Cabral et al. (2022), which studies rationales for workers' compensation mandates using Texas. Ultimately, they find no evidence for a mandate coming from adverse selection, market power, or externalities. The authors use a revealed preference approach to quantify selection, similar to Einav et al. (2010), which requires an assumption that an employee's value from coverage flows through the employer's choice (which they justify using Gruber and Krueger (1991)). This implies that employees must be fully aware of their employer's coverage choices and that their employment decision properly reveals their value of workers' compensation benefits.⁵ My analysis focuses on employee outcomes to better understand if seminal compensating differentials work from a mandated state computed based on marginal benefit changes (Gruber and Krueger, 1991) speaks to the experience of employees losing their coverage completely.

Finally, this work contributes to the broader literature on the use of insurance coverage mandates in public policy, which is directly related to how we think choice plays a role in

⁴The same authors also provide reviews of other components of workers' compensation markets and provide an in-depth account of the history of workers' compensation in the US (Fishback and Kantor, 1996, 1998).

⁵Smith (1979) describes the ambiguity in using compensating differentials in settings where the labor may not be functioning properly, such as employee miscalculation of risk or lack of workers' compensation salience. Moss (2002) agrees and suggests that individual misunderstanding of risk is in and of itself a rationale for government involvement in risk management.

optimal insurance settings, an application of choice in policy-making more broadly (Chetty, 2015; Bhargava and Loewenstein, 2015; Viscusi and Gayer, 2015). Coverage mandates, which are inspired by the (lack of) role of choice, have been widely studied in health insurance⁶ (Gruber and Madrian, 1994; Auerbach et al., 2010; Kolstad and Kowalski, 2016; Barkowski et al., 2020; Richards and Tello-Trillo, 2021; Barkowski and McLaughlin, 2022; Maclean and Webber, 2022), but little empirical evidence exists in other insurance settings (Wriggins (2012) discusses auto insurance). Thus, my analysis builds past Cabral et al. (2022) to ask about the consequences of choice on individuals.

Given the history of workers’ compensation and the nature of the tradeoffs faced by employers and employees, Section 2 describes the institutional background. Section 3 describes the data, while Section 4 presents the empirical methodology. Sections 5 and 6 present results and conclude, respectively.

2 Institutional Background

Workers’ compensation was the first social insurance program in the US.⁷ As a state-based system, the first workers’ compensation programs were created starting in 1911, long before federal social insurance programs were created. Each state has its own workers’

⁶The papers cited here is not an exhaustive list, but they provide reviews of the space and results that show coverage mandates are relevant to individual outcomes beyond coverage itself.

⁷It is important to define social insurance in this context, given several definitions have been proposed. The National Academy of Social Insurance defines social insurance as: “Social insurance encompasses broad-based systems that help workers and their families pool risks to avoid loss of income due to retirement, death, disability, or unemployment, and to ensure access to health care” (via the Academy’s website: <https://www.nasi.org/about/>). This definition may be problematic given it doesn’t rule out transfers of resources across individuals. Chetty and Finkelstein (2013) recognize that and define social as insurance as “...government programs that transfer resources across states of nature after an individual is born rather than transfers of resources across individuals (e.g., through re-distributive taxation).” This definition is adequate but would not exclude payments from programs like the National Flood Insurance Program or Terrorism Risk and Insurance Program, which are typically not considered as social insurance programs. Krueger and Meyer (2002) offer a more refined version and define it as “...compulsory, contributory government programs that provide benefits to individuals if certain conditions are met.” This definition would exclude transfer programs like Medicaid and Supplemental Security Income, while at the same time suggesting that optional workers’ compensation programs are not social insurance. This nicely highlights the importance of this study in understanding more about deviating from social insurance principles. The definition I proffer and use as a guide combines the best elements of these to suggest that *social insurance programs are programs which, with government intervention, provide statutorily defined benefits to workers via a compulsory program that is funded by contributions of the employer, employee, or both and exists to transfer resources across states of nature and not across individuals.*

compensation insurance market, sets its own benefits levels by statute, and makes its own decisions regarding whether to make coverage compulsory. In each state except for Texas, workers' compensation coverage is required for employers to provide to their employees, which has been true for the last several decades.⁸ In this setting, the employer organizes the coverage arrangement and every employee within a state receives the same protection.

Before workers' compensation programs were created, employees were quite likely to lose lawsuits for injuries at work, and in these times, a lawsuit against one's employer was the only way to be compensated for one's injury costs. This was generally due to the fact that employers could use common law defenses to win against most employee suits. These defenses include: 1) claiming the employee may have caused the accident on his/her own, 2) claiming a fellow employee was at fault, and/or 3) claiming that the injured employee had "assumed the risk" of injury by accepting the offer of employment. In the early 1900s, even before workers' compensation programs were created, states generally moved to eliminate these defenses for employers.

When workers' compensation programs were eventually created, they were largely considered the "grand bargain." This bargain refers to items given up by both employers and employees to design the workers' compensation system with two key pillars: 1) employees are given near guaranteed payment for injuries at work in the form of lost income replacement (not full) and medical care; 2) employers are generally protected from employees filing lawsuits for larger damage amounts. On the first element, as long as employee can show that a work injury was an accident arising out, and in the course, of employment, then they will be paid. Employers, on the other hand, gave up the ability to robustly reject claims, even if the employee made the accident substantially more likely. On the second element, an employee is not able to choose between workers' compensation benefits and suing his or her employer for negligence. This makes workers' compensation the only remedy available for work injuries, which forces the employee to give up their chance to win more in court.

⁸Wyoming and South Dakota are known to allow for limited opt-outs compared to Texas but more so than other states, where it is common only for very small firms to opt out.

Insurers also joined in on this bargain, happy to use coverage mandates as a way to expand their coverage options (Fishback and Kantor, 1998). Not all scholars agree, however, that all three parties were eager to make this bargain. Weinstein (1967), for example, suggests that many employers became eager to force the workers' compensation system on employees after states removed the common law defenses described earlier, primarily to stem a burgeoning of employee tort success. This dynamic has been recognized by others (Burton and Berkowitz, 1971) and has been the impetus for deep questions about whether workers are really equal winners with this bargain in the present day (Amchan, 1983; Spieler, 2017).

In Texas, employers have the unique choice about whether to be covered or uncovered with respect to this system, and the employee chooses where to work (assuming to be maximizing over a decision set that includes this information).⁹ Employees do not have access to workers' compensation insurance as a product to buy themselves. Coverage, in Texas and every state, can be obtained through the purchase of insurance or the arrangement of a certified self-insurance scheme (to be approved by the state for financial soundness).¹⁰ Either of these arrangements is required to pay benefits to injured workers as prescribed by the statutory benefit levels in a given state, which may change over time. When an employer opts to be uncovered in Texas, they are also known to be a "non-subscriber," which refers to the fact that uncovered employers are not subscribed to the "grand bargain" discussed above, meaning the employee may file a tort suit against the employer for more than what subscribers are mandated to provide per statute. The employee could also lose that suit and receive less (down to zero) than statutory levels.

A caveat, however, does exist in that uncovered employers may still provide alternative injury plans (broadly governed by the Employee Retirement Income Security Act of 1974,

⁹Interestingly, section 406.034 of the Texas Labor Code gives an employee the option to opt out of workers' compensation coverage at their employer and retain the right to pursue negligence claims. Employees must elect this, in writing, within 5 days of being covered (either from the hire date or the date of coverage if the firm is newly subscribed). This only exists as an option for employees at subscribed firms. Data as to this option remains unavailable via open records request.

¹⁰Ohio and Wyoming are the only two states that do not allow for self-insurance; these states also operate monopolized insurers that provide virtually all of the workers' compensation insurance to their employers.

or ERISA for short) to their employees for on-the-job injuries. The Division of Workers' Compensation (DWC) at the Texas Department of Insurance (TDI) releases reports about non-subscription, with evidence that of the 25% of employers that were non-subscribers in 2022 (with nearly 1.5 million employees), 30% provided an alternative injury plan ([DWC, 2022](#)). This implies that 17.5% of Texas employers (roughly 5% of employees, nearly 500,000 workers) operated in an environment where the *only* compensation mechanism for work injuries was through a lawsuit. The percentage of non-subscribers offering alternative injury plans has declined steadily, from 56% in 2001 to 33% by 2012.

When they are offered, these plans tend to be more strict, which is in keeping with the evidence from [Jinks et al. \(2020\)](#) that non-subscription is a cost-cutting endeavor. Strictness may mean several things. First, covered services are more limited. While most workers' compensation systems in Texas cover acupuncture and chiropractor services (claim instructions are even readily available online for the largest insurer: Texas Mutual), only 20% and 13% of alternative plans cover these, respectively ([DWC, 2022](#)). Second, reporting requirements are much more strict. Among subscribers in Texas, employees generally have 30 days to report an injury; in 2022, 57% of non-subscribers require same day reporting with another 32% requiring reporting within 24 hours or by the next day. Third, and finally, benefits are more limited. For instance, among the 30% of non-subscribers who offer alternative plans, only 67% pay medical benefits for as long as is necessary and only 47% pay lost income replacement benefits. Death benefits are always paid to a widow(er) and dependent children among Texas subscribers; yet, among non-subscribers, 71% offer death benefits to the spouse and just 27% to dependent kids.

Thus, in this non-subscriber world, employees are forced to manage substantially more risk, both by taking more limited benefits with less-uniform rules (that cannot be easily searched as statewide rules, for example) and through navigating potential tort claims (and hiring necessary legal counsel). Employers, too, could face more risk if the lawsuit avenue is 1) salient and 2) used efficiently. One concern in this setting is the use of binding arbitration

agreements, which have become more prominent and increasingly upheld by courts ([Sargent et al., 2020](#)). In 2020, 15% of non-subscribers used these to force an employee into arbitration rather than allowing them to pursue a negligence claim in court ([DWC, 2020](#)). 32% of non-subscribers do not pay benefits at all to employees who do not agree to binding arbitration, while 37% appear to make it a condition of employment; 35% of non-subscribers require employees pay all or some of the eventual arbitration costs. These descriptive facts highlight additional potential concerns within the system for workers, but they also highlight that negligence suits remain open to many of the employees who are employed by non-subscribers, which is relevant to the calculus of costs, benefits, and salience.

3 Data

This project combines data from several sources and types of acquisition. The unique administrative data comes from several open records requests in Texas, four of which were fulfilled by TDI and one by DWC.¹¹ I have also gathered publicly-available data from the Quarterly Census of Employment and Wages (QCEW), Bureau of Economic Analysis (BEA), Census Bureau, Bureau of Labor Statistics' (BLS) Survey of Occupational Injuries and Illnesses (SOII), Texas Office of Court Administration (OCA), and the Texas Secretary of State's Election Archive. I also have access to data on personal injury law firm advertisements by designated market area (DMA) - of which there are 20 in Texas - through Nielsen's Ad Intel product provided by the University of Chicago's Booth School of Business.

3.1 Open Records Data

The open records data was provided in five separate requests. Two requests focus on administrative reporting of covered and uncovered firms. Texas makes covered employer information public online to promote transparency, and the proof-of-coverage data I requested reports this information historically. This includes name of firm (and establishment at a finer level), policy effective dates, insurer name, industry, address, and some risk rating in-

¹¹The author is willing to share resources on duplicating such open records requests, as was done by [Cabral et al. \(2022\)](#). I intend to build a public dataset out of these open records requests in the coming years, which I can make available online and by request once organized. Lawyers on staff at TDI have reassured me that I may do so.

formation. One weakness of this data is that it is not linked across time within a firm or establishment. Similar information exists for uncovered firms, which must file a non-covered form with the state each year, providing me with names, addresses, and industries. Firms must also report when employees were notified of the uncovered status relative to the effective date of non-coverage. Both sets of this information - for covered and uncovered firms - is available over the 2010-2022 period.

Two of the other three requests focused on injury information. The most substantial of those is unit statistical data, which provides injury-level data among covered firms over the 2015-2021 period (which is similar data to [Cabral et al. \(2022\)](#) but over a different period). These data are linked across employers over time but remain randomized to protect employee identities. The data include too many items to list here, but most relevant are, for each injury, information related to dates of injury, type of injury, costs expended on the claim (separated by medical and lost income), settlement information, and fraud identifiers. The nature of this dataset is that it is insurer-relevant information, so information on the firm as a whole (payroll, premiums, industry, location, etc.) is observable as are loss statistics for all injuries within a firm. Despite needing to file with the state for each injury, this data is not available to this level of granularity for uncovered firms via an open records request. TDI was willing, however, to release summary statistics from uncovered firm injuries, which provide injury counts by industry, county, gender, race/ethnicity, and various injury details.

The final open records request was fulfilled by DWC and includes historical information from the Occupational Employment and Wage Statistics (OEWS) program that was unavailable publicly from either BLS or the Texas Labor Market Information (LMI) portal. These data provide detailed wage information, such as wages at the mean, entry level, experienced level, and 10/25/50/75/90 percentiles. It is further split by hourly and annual rates, industry, and metropolitan statistical area (MSA).

3.2 Other Data

3.2.1 Labor Market and Demographics

Without firm-level data, most of the data discussed above - especially information about rates of uncovered and covered firms - must be aggregated to the industry-county level. With all 254 Texas counties visible in the data, and labor market information for over three hundred North American Industrial Classification System (NAICS) codes, there is still enough power to describe these relationships. To build out labor market information at the industry-county observation level, the QCEW provides information related to total number of establishments (to create an uncovered rate, for example), total annual wages (including taxable wage and contribution values), and - more importantly - average weekly and annual wages that adjust for the number of employees and average work time.

Given the aggregation to the industry-county level and conditional on the specifications I present below, additional information about the changing dynamics within a county is quite important. To support that idea, I gather county-level GDP data from the BEA and county-level population data from Census, both over the same period as the open record coverage requests and QCEW data (2010-2022). To provide summary information about which industries tend to be the riskiest, I use data from the BLS' SOII to identify injury incidence rates across industries over time, which break incidence rates down by cases with time spent away from work and cases with job changes or restrictions.

3.2.2 Legal Information

I also make use of data from the Texas court system. First, I can collect information from OCA that details - by year and county - the number of civil cases (broad category for work injury suits brought against employers) with case statuses and the same information but for a more specific category of civil case denoted "other injury or damage," which separates personal injury claims from cases related to motor vehicles, medical malpractice, other professional liability, asbestos, other product liability, real property, and contracts.

I can also locate data related to whether there were mistrials declared, seated juries, and self-representation among the broad civil case category.

As described below in Section 4, I make use of Texas district court (which has jurisdiction in personal injury cases such as cases from workers against uncovered employers) election data, primarily because Texas has competitive, partisan elections and there are nearly 500 district courts across the state. To fully capture the role of incumbent parties and judges over the 2010-2022 period, I hand-collect historical election data from the Texas Secretary of State’s archive for all elections since 2002. With judges serving four year terms, a judge on the bench in 2010 was elected in 2006, and to be sure about their incumbent status, going back to 2002 is necessary. The idea with these data is to identify county-year observations that are operating under certain parties and - most importantly - new judges (who may also be from new parties) compared to the prior status quo.

4 Empirical Methodology

Fundamentally, this paper seeks to estimate the relationship between the presence of uncovered establishments in a given industry-county observation with the employee outcomes outlined above. One way to summarize that relationship is by suggesting the following specification for wages:

$$Wages_{i,c,t} = \alpha + \beta UncoveredRate_{i,c,t} + \gamma_i * \delta_c + \lambda_t + \epsilon_{i,c,t}, \quad (1)$$

where *Wages* can be total annual wages, or an average weekly or annual wage that controls for employment levels (all provided by the QCEW), in industry *i*, county *c*, and year *t*. Traditional economic theory would suggest that β should be positive, where when more employers become uncovered, wages rise to compensate. There are several reasons, however, why the final relationship may not be different from zero: workers’ compensation coverage is not salient, the outside option (and associated risk) of suing an uncovered employer is misunderstood, and/or the value of workers’ compensation benefits is less than expected. The dependent variable can be replaced with lawsuits and injuries, as well.

Using the statutorily-required filings of uncovered firms, which includes all of the establishments to which those filings apply, to aggregate to the industry-county level with a measure of uncovered establishments is useful because it can be matched up with QCEW information about total establishments at the same level. Given this data has not been used yet in the academic literature and how important the measurement of uncovered firms is to the analysis, it seems worthwhile to describe in detail. Then, I consider endogeneity concerns related to Equation 1 and propose an instrument.

4.1 Measuring the Presence of Uncovered Establishments

Regularly uncovered firms in Texas are required to file DWC Form 005 each year, and newly-uncovered firms must do so whenever they terminate coverage (and annually thereafter).¹² Via open records request, TDI was willing to share responsive information from most of the form over the 2010-2022 period, including de-identified firm numbers (to track parent firms over time), filing identifiers (to track all establishment filings under a parent firm in a given year), filing dates, establishment name and address, and then an NAICS code. In the raw data, there are more than 770,000 observations.

After merging the ZIP codes with a county mapping in Texas, nearly every observation reported valid zip codes. For the remaining 890 observations, I merge based on city name (using mappings from the Texas Department of Transportation), correct some obvious city misspellings, drop parent firm filings not based in Texas, and can correctly match nearly every observation. 102 observations are dropped in the end for addresses, cities, and ZIP codes that could not be corrected or imputed.

I define an industry as being based on four digit NAICS codes, even if firms report six digit codes. This is the approach for several reasons: 1) some firms only report four digit codes and would need to be dropped if using six, 2) six digit NAICS codes eliminate some ability of the QCEW to report wages and earnings, and 3) there is a chance that the fifth and six digits could be measured with error (maybe more so than others). Some NAICS codes

¹²Readers may access the form here: <https://www.tdi.texas.gov/forms/form20employer.html>.

changed in 2012, 2017, and 2022 based on regular NAICS updates, so I manually update those NAICS codes for the applicable years.¹³ If an industry code is reported as undefined (“UNDE”), I first search for the modal industry code within a firm and replace it; if an entirety of a firm’s responses have an undefined NAICS code, they are dropped.

Some additional cleaning is necessary to bring establishment coverage data in line with how the QCEW measures establishments. Namely, duplicates need to be removed from the data (places with the same parent identifier but who also have a matching address with another establishment in the same industry, county, and year).¹⁴ This ensures that establishments are properly lining up with the QCEW. For the main analysis, I also drop the parents that report having establishments, because the parent firm is generally listed as a P.O. Box address and is not a place of business. An exception is to keep parent firms that do not report having establishments, as that makes it one place of business. An example for the reader to contextualize why this is an important step to ensure the QCEW total establishments information is valid: many health centers (parent firms) file separate forms for each department in the one physical location, maybe with different addresses for mailing purposes, but ought to only count as one establishment.¹⁵

These cleaning steps ultimately result in a sample size of just over 450,000 establishment-year observations which collapse to over 80,000 industry-county-year observations. There are just under 7,000 observations that are dropped for not matching between the coverage data and the QCEW, which likely implies some incorrect reporting of (otherwise valid) ZIP and NAICS codes. In addition, some industry-county-year observations employ too few people to report wage information at all, so those are dropped from the analysis. The final sample,

¹³For example, NAICS codes for “warehouse clubs, supercenters and other general merchandise retailers” changed from 4523 to 4552 in 2022, but many firms in this classification did not update their old code from 2021 to 2022.

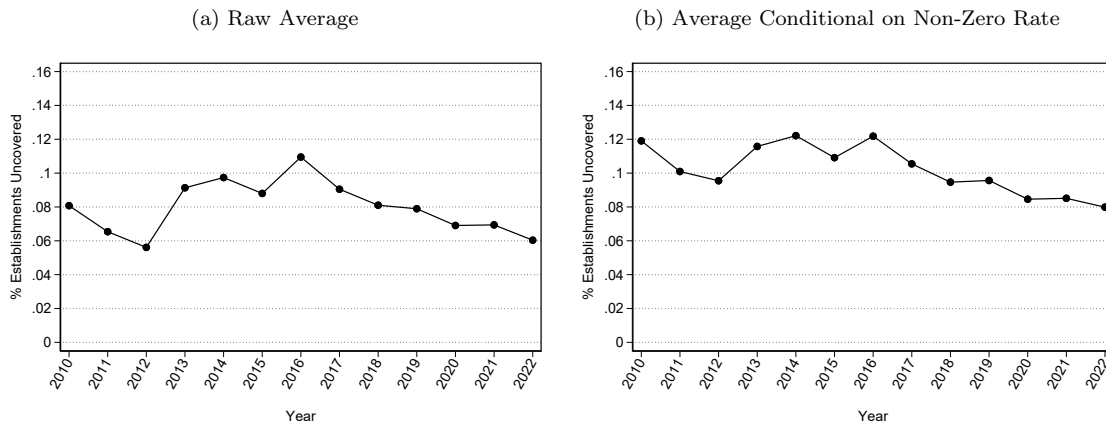
¹⁴Firms and establishments do not always report exactly the same type of address each year, thus I also drop likely duplicates where large sections of the address are identical within a year, industry, and parent identifier.

¹⁵In addition, at the very end of the cleaning process, I will drop the (very few) cases where the number of uncovered establishments exceeds the total number of establishments in the QCEW. This represents less than 0.05% of all observations.

including all industry-county-year observations with an establishment and reported wages, numbers roughly 162,000.

Defining $UncoveredRate = \frac{UncoveredEstablishments}{TotalEstablishments}$, Figure 1 shows the weighted (by total employment in an industry-county-year) mean of the uncovered rate over time, raw in panel (a) and among those observations with a non-zero uncovered rate in panel (b). While my rates do not exactly match those of TDI from their surveys, the patterns match fairly well to their reports of the percentage of employers who are non-subscribers (oscillating between 17-20% between 2010 and 2022), with similar responsiveness in various years and a general decline since 2010 (panel (b) depicts the most strikingly similar patterns). Appendix Figure A.1 provides the analog figure without weights. As an alternative to the uncovered rate, the analyses in the next section will also use indicator variables to track if a given industry-county has experienced an increase in its uncovered rate. This focuses more on the effects in one direction that is most relevant to the legal and economic tradeoffs in this system.

Figure 1: Uncovered Rates in the Data



Notes: This figure displays uncovered rates over time, for all industry-county observations in panel (a) and among a smaller subset in panel (b) that had a non-zero uncovered rate. Size of total employment in an industry-county observation used as the weight. Sources: Texas open records requests and QCEW, 2010-2022.

One version of the uncovered rate that is not reported publicly is uncovered rates by county, which can then be matched up with population estimates from Census. Figures 2 and 3 present uncovered rate and population quartiles by county over time, respectively.

Appendix Figure A.2 mimics Figure 2 but uses raw uncovered rates instead. These maps generally indicate that uncovered rate quartiles map to population quite well, where urban areas see more uncovered rates. There are some notable exceptions in West Texas, however, where some more rural areas have the highest rates of uncovered employment. Counties in white are missing in a given year because they are small enough counties where disclosure issues prevent seeing the wages in their industries (i.e., Loving County had 84 total residents in 2010).

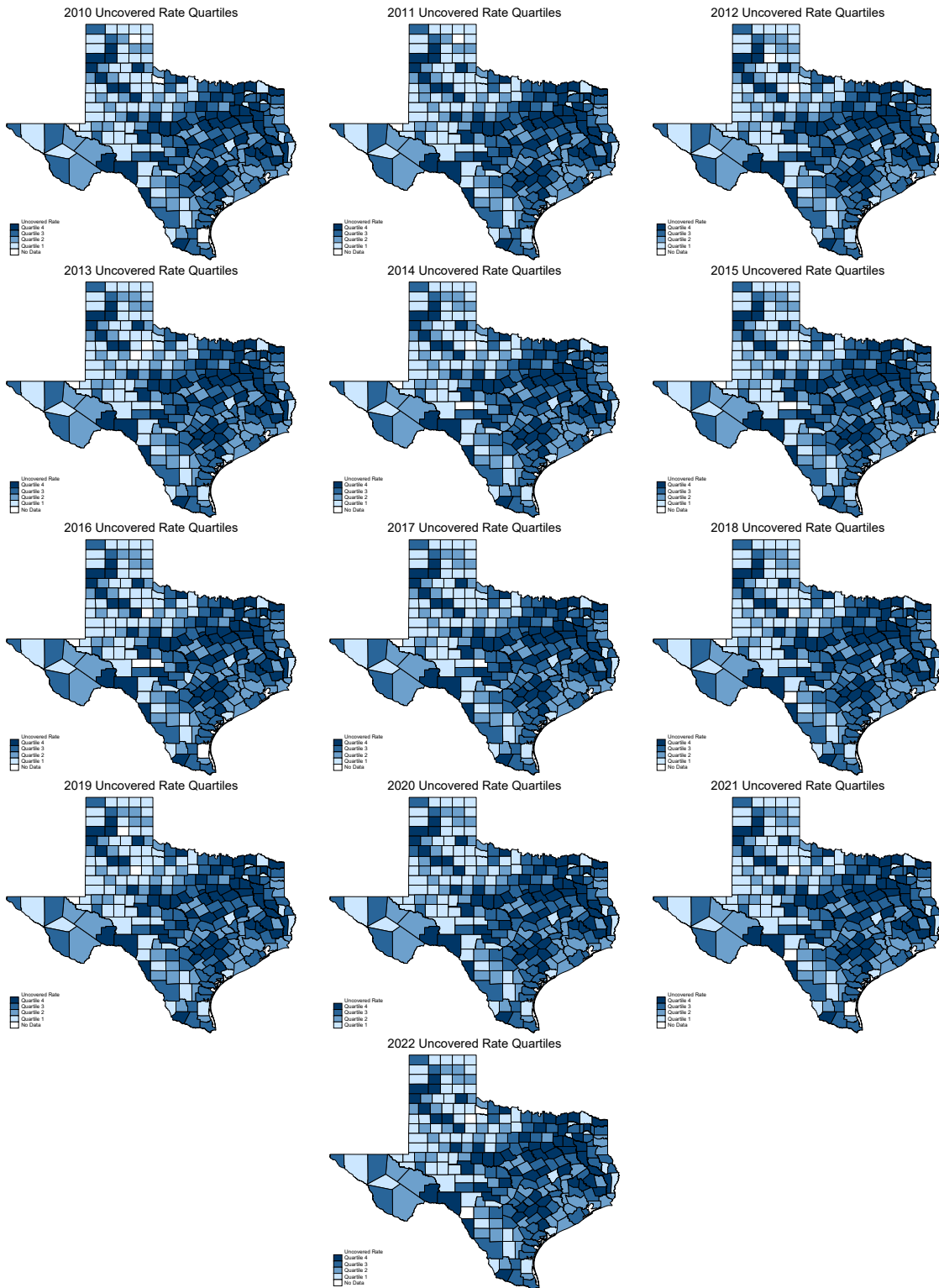
4.2 Endogeneity Concerns

There is a concern with Equation 1, however, in that the uncovered rate may be endogenous, especially when this paper aggregates from the firm-level. Several items may affect both wages and the uncovered rate. For example, higher injury rates (+) at a firm may mean a smaller chance of being uncovered (-) but a higher wage to compensate for the risk (+). This may suggest that there is negative omitted variable bias, where estimating Equation 1 would lead to inference based on a negative β . The omitted variable bias could also be positive, if more injuries leads to higher insurance premiums and thus a push to being uncovered. The evidence provided by Jinks et al. (2020), the most closely-related paper to this question, provides mixed evidence but suggests that being uncovered comes with more denied claims, even under alternative plans. Ultimately, this dynamic is an open empirical question (to which I hope to provide evidence in the near future). TDI suggests that 23% of uncovered firms report their reason for being a non-subscriber to be that they had few injuries; yet, 17% said premiums were too high, which likely reflects - to some extent - past injury history (DWC, 2022).

4.2.1 Proposing an Instrument

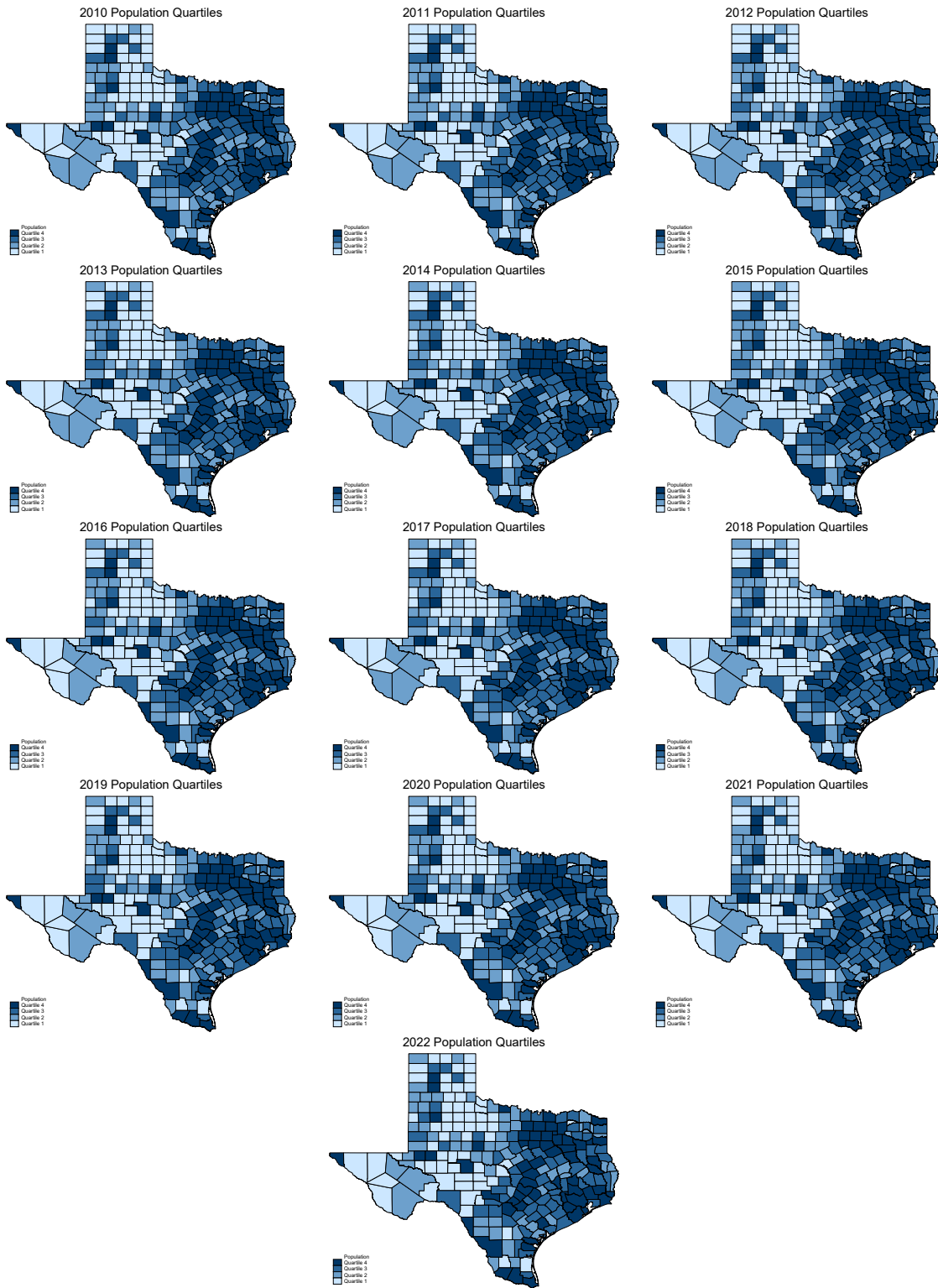
Using a valid instrument presents two key advantages in this setting (applied from any other): 1) it can bolster causal inference claims related to the effect of not being covered by workers' compensation, which could be a relevant parameter for policymakers, and 2)

Figure 2: Uncovered Rate Quartiles Across Counties by Year



Notes: These maps display uncovered rate quartiles by county. The darker the blue, the higher the uncovered rate quartile. Sources: Texas open records requests and QCEW, 2010-2022.

Figure 3: Population Quartile Across Counties by Year



Notes: These maps display population quartiles by county. The darker the blue, the higher the population.
Sources: Census, 2010-2022.

concerns about measurement error in the uncovered rate may be eased, especially if a reader is worried about the idea that certain firms may misreport or make mistakes that could also be correlated with their wages.

To identify an instrument, it is helpful to identify *why* an employer opts out (while this may be hand-wavy now, additional empirical support can be provided in the future). This is a non-exhaustive list, but it serves as a jumping-off point for an instrument's consideration.

- Insurance is too expensive, either because it is not perfectly experience-rated¹⁶ (and an employer feels they are subsidizing unsafe competitors) or because it is experience-rated properly, leading to prohibitive premiums.
- An alternative injury plan may be cheaper than insurance, which is in part related to the item just above but also separate in that the motivation may not be premiums but the ability to be more restrictive.
- Employees are unaware of, or miscalculating, their tort options and may be nudged (forced) to settle or pursue arbitration.
- The court system becomes friendlier to employers, decreasing the expected loss associated with being uncovered, either in terms of severity (damages awarded) or frequency (number of cases moving forward).

The only item in that list that is easy to measure and not endogenous itself is the final item. Insurance costs are likely passed through to wages (Gruber and Krueger, 1991), as would be things related to injuries. Absent a survey of how (if) individuals understand their employers' coverage choice and subsequent tort remedies, the fourth motivation suggests changes in court systems ought to influence both 1) the decision for employers to be uncovered and 2) the experience of uncovered employees.

¹⁶In the workers' compensation context, experience-rating is pervasive. This technique means that premiums reflect an employer's injury history.

4.2.2 Texas' Judicial Elections

Texas is home to over 450 state district courts, which tend to hold jurisdiction over large civil cases, such as personal injury suits from an injured employee against an uncovered employer. Judges run in partisan elections every four years. Using hand-recorded data from all district judge elections between 2002-2020, I am able to determine, for all counties, 1) the party of their judge(s), 2) whether or not they are being served by an incumbent judge, and 3) whether or not they are being served by an incumbent party. No ex-ante position is necessary on which party's judges would be more or less favorable to workers.

In theory, a change in judge (induced by either retirement or an election loss) and a change in party (either by defeating an incumbent or winning an open seat) are both relevant to the calculus for being uncovered, as described above, and plausibly exogenous with respect to wages. It is unlikely, for instance, that a judge election changes wages other than through changing the propensity employers served by a given judge opt to be uncovered. To visualize, we can imagine the first stage of a 2SLS specification as:

$$\widehat{Uncovered}_{i,c,t} = \alpha + \beta CourtChange_{c,t} + \gamma_i * \delta_c + \lambda_t + \epsilon_{i,c,t}, \quad (2)$$

where *Uncovered* can either be the continuous rate or an indicator for an increase in the rate. *CourtChange* can be any number of indicators to be tested as instruments in the next section, but most focus will be given to new judge and new party percentages (given some counties have several district courts). The second stage is as follows:

$$Wages_{i,c,t} = \pi + \theta \widehat{Uncovered}_{i,c,t} + \gamma_i * \delta_c + \lambda_t + \epsilon_{i,c,t}. \quad (3)$$

The next section presents results, with respect to the endogenous OLS specification and the first stage of the 2SLS specification (the second stage will be added after feedback on the instrument itself). Standard errors will be clustered at the industry-county level, i.e. at the level of observation in the panel, which yields more than 20,000 clusters. (Note to the

committee: My most immediate plan is to spend more time on descriptive evidence of the first stage, and so I ultimately intend to make much of my results section descriptive, both for the IV but also for other places it seems best).

5 Results

The results from estimating the plausibly endogenous specification from Equation 1 are presented in Table 1, with the continuous version of the uncovered rate used in Panel A and an indicator for having an increased uncovered rate (relative to last year) used in Panel B. The results in both panels are largely consistent with one another, though the coefficient sizes in Panel B are significantly smaller than in Panel A. Across earnings categories, the uncovered rate is negatively associated with wages. Given the use of industry-by-county fixed effects, we can think about these coefficients representing changes in the uncovered rate and the associated changes in earnings (positively or negatively). Panel B specifically asks about industry-county observations where the rate is increasing.

Table 1: Uncovered Rates and Earnings

	(1) Total Annual Wage	(2) Total Annual Taxable Wage	(3) Total Annual Contributions	(4) Average Weekly Wage	(5) Average Annual Wage
Panel A					
Uncovered Rate	-1820.2*** (334.9)	-185.5** (72.52)	-1.793 (2.154)	-16.91** (7.072)	-878.9** (367.7)
Reference Mean	10,805.77	2,199.12	52.86	935.41	48,646.30
Panel B					
Increased Uncovered Rate	-213.9*** (61.22)	-40.25*** (11.73)	-0.0105 (0.380)	-2.339** (1.030)	-121.6** (53.57)
Reference Mean	17,360.10	3,766.00	84.05	475.50	48,135.49
Fixed Effects					
Industry-by-County	✓	✓	✓	✓	✓
Year	✓	✓	✓	✓	✓
Observations	159201	159201	159201	159201	159201

Notes: Results from estimating Equation 1. Columns (1)-(3) are in thousands of dollars. Variables measured in 2022 US dollars. Standard errors clustered at the industry-county level. Reference mean refers to the average of the outcome in industry-county observations with an uncovered rate of zero (Panel A) and no increase in the uncovered rate (Panel B). All outcome variables were winsorized at the 5/95 percentiles. Source: Texas open records requests and QCEW, 2010-2022. Statistical significance is given by: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Thus, the smaller coefficients in Panel B may well be explained by the fact that there is less mixing of uncovered rates in both directions. In general, though, the specification seems to be quite precise, even with some economically small coefficients. Panel A suggests, for example, that moving from an industry-county with no uncovered establishments to one where they are all uncovered is associated with an \$878 reduction in individual's annual wage. The reduction is much smaller in Panel B. The results from columns (1) and (2) suggest that total annual (taxable) wages paid decreased in a larger fashion (as a percentage of the reference mean) than employment-adjusted wages in columns (4) and (5), which could indicate that employment decreased by more than wages. The decreases in Panel B tend to be similar in a relative sense, which suggests less of an employment effect and more of a direct wage effect. The negative coefficients across the board are at odds with the idea that workers fully understand and internalize the loss of coverage, both in terms of optimal wages and the outside legal option.

To better understand the role endogeneity (and measurement error) may be playing negative results, especially given negative omitted variable bias was suggested above, I now turn to proposing judicial election instruments. To understand the strength of various instruments, Table 2 present first stage estimates from estimating Equation 2 on various judicial items: the percent of a county's district judges that are Republican, are new judges, are of a new party, and had been elected in a competitive race (i.e., a winning percentage of less than 100). There is one additional outcome which is the average winning margin (which runs between 50.01 and 100). I use both potential endogenous regressors, though the indicator for an increased uncovered rate is theoretically better, because it avoids ambiguous cancelling out of positive and negative uncovered rate changes. While I take no position on the expected relationship between party and the uncovered rate, it is expected that - if the rationale laid out above holds - more new judges, more new parties, and more competitive elections should be negatively related to uncovered rates and especially increasing rates. This is because these variables indicate places with more volatile judicial systems where an

employer cannot rely on stability in personal injury suits. The winning percentage should be positively related to both outcomes, given a higher value there means less volatility and more security for uncovered employers.

Columns (1) and (6) indicate no explanatory role of a judge's party on (increases in) the uncovered rate. This is not all that surprising, given that the instrument proposed here is more about disruptions to the sense of security an employer feels about being uncovered. Among the other variables, there are consistent signs between dependent variables, which all match expectation, though they come with varying levels of statistical significance. Generally speaking, there are no strong relationships using the raw, continuous uncovered rate. Given there are many zeroes in the uncovered rate in its raw form, and because the uncovered rate changes identified by the fixed effects design could be positive or negative, the indicator for a positive change in the uncovered rate better isolates the theory proposed.

Columns (7)-(9) identify three potentially strong instruments for the uncovered rate. In these cases, new judges, new parties, and more competitive elections are statistically significant in their relationship with increasing uncovered rates. Considering each specification, the new judge and new party cases produce F -statistics of 49.44 and 66.44, respectively. The competitive race is statistically significant but not overly strong when considering the first stage altogether ($F = 6.06$). The same is true for the winning margin in both sets of outcomes, with an F -statistic of just under four. Overall, these results indicate the chance for strong instruments that - importantly - match the expected relationship with the endogenous regressor of interest.

6 Conclusion

Motivated by the recent desire in several states to remove their mandate for employers to purchase workers' compensation coverage, I use the optional Texas system, which is unique in the US, to provide the first evidence - to my knowledge - on employee labor market and legal outcomes. The work is relevant both to the workers' compensation space specifically and the role of insurance coverage mandates as policy solutions more broadly. I take advantage

Table 2: First Stage Estimates

	Uncovered Rate				Increased Uncovered Rate					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
% Republican	0.000249 (0.00170)					-0.00280 (0.00477)				
% New Judge		-0.000136 (0.00109)					-0.0226*** (0.00322)			
% New Party			-0.00248* (0.00143)					-0.0350*** (0.00429)		
% Competitive				-0.00297* (0.00167)					-0.0128** (0.00521)	
Avg. Winning Margin					0.0000801* (0.0000414)					0.000245* (0.000130)
Industry-by-County	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Year	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	159201	159201	159201	159201	159201	159201	159201	159201	159201	159201

Notes: Results from estimating Equation 2. Columns (1)-(5) use the continuous uncovered rate as the endogenous regressor while columns (6)-(10) use a binary indicator for an increase in the uncovered rate. Standard errors clustered at the industry-county level. Source: Texas open records requests and QCEW, 2010-2022; Texas election archive, 2002-2020. Statistical significance is given by: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

of a unique administrative dataset, constructed by several open records requests in Texas, that allows me to identify the rates of uncovered establishments at the industry-county level and, therefore, match to publicly-available earnings and wage data.

First, I show that - in opposition to what economic theory (namely, that of compensating differentials) might suggest - the rate of uninsured employers in a given industry-county observations is negatively related to both total annual wages (mixing employment and wage effects) and average wages at the weekly and annual levels (employment-adjusted). While not all of the results are economically significant, they are statistically precise, which allows me to rule out meaningful positive effects. I then suggest that one reason we may see that becoming uninsured is related to lower wages (i.e., employees are not compensated for taking fewer benefits and more risk) comes from an endogeneity issue caused by, for example, unobserved firm injury risk.

I then suggest two instruments related to partisan judicial elections in Texas. Judge elections are theoretically relevant to the firm's coverage decision because an uncovered employer opens itself to the chance of being sued by employees for a work injury (which is not possible if the employer is covered). Using hand-collected election data across Texas' more than 450 district courts between 2002-2020, I document that more volatile judge settings (i.e., counties that operate with more new judges or judges of new parties) are related to fewer industry-county observations increasing the rate of uncovered employers, presumably because employers sense more risk from employee lawsuits. I then propose using these instruments to estimate 2SLS to understand if the negative effects previously described are due to endogeneity or measurement error in the uncovered rate.

References

- Amchan, A. J. (1983). “Callous Disregard” for Employee Safety: The Exclusivity of the Workers’ Compensation Remedy Against Employers. *Labor Law Journal* 34(11), 683–696.
- Auerbach, D., J. Holtzblatt, P. Jacobs, A. Minicozzi, P. Moomau, and C. White (2010). Will Health Insurance Mandates Increase Coverage? Synthesizing Perspectives from Health, Tax, and Behavioral Economics. *National Tax Journal* 63(4, Part 1), 659–680.
- Barkowski, S. and J. S. McLaughlin (2022). In Sickness and in Health: Interaction Effects of State and Federal Health Insurance Coverage Mandates on Marriage of Young Adults. *Journal of Human Resources* 57(2), 637–688.
- Barkowski, S., J. S. McLaughlin, and A. Ray (2020). A Reevaluation of the Effects of State and ACA Dependent Coverage Mandates on Health Insurance Coverage. *Journal of Policy Analysis and Management* 39(3), 629–663.
- Bhargava, S. and G. Loewenstein (2015). Behavioral Economics and Public Policy 102: Beyond Nudging. *American Economic Review: Papers & Proceedings* 105(5), 396–401.
- Brown, C. (1980). Equalizing Differences in the Labor Market. *The Quarterly Journal of Economics* 94(1), 113–134.
- Burton, Jr., J. F. and M. Berkowitz (1971). Objectives Other Than Income Maintenance for Workmen’s Compensation. *Journal of Risk and Insurance* 38(3), 343–355.
- Butler, R. J. (1996). Lost Injury Days: Moral Hazard Differences between Tort and Workers’ Compensation. *Journal of Risk and Insurance* 63(3), 405–433.
- Cabral, M., C. Cui, and M. Dworsky (2022). The Demand for Insurance and Rationale for a Mandate: Evidence from Workers’ Compensation Insurance. *American Economic Review* 112(5), 1621–1668.
- Chetty, R. (2015). Behavioral Economics and Public Policy: A Pragmatic Perspective. *American Economic Review: Papers & Proceedings* 105(5), 1–33.
- Chetty, R. and A. Finkelstein (2013). Social Insurance: Connecting Theory to Data. In A. J. Auerbach, R. Chetty, M. Feldstein, and E. Saez (Eds.), *Handbook of Public Economics*, Volume 5, pp. 111–193. Elsevier.
- DWC (2020). Biennial Report to the 87th Texas Legislature. Technical report, Division of Workers’ Compensation, Texas Department of Insurance, Austin, TX.
- DWC (2022). Employer Participation in the Texas Workers’ Compensation System: 2022 Estimates. Technical report, Division of Workers’ Compensation, Texas Department of Insurance, Austin, TX.
- Einav, L., A. Finkelstein, and M. R. Cullen (2010). Estimating Welfare in Insurance Markets Using Variation in Prices. *The Quarterly Journal of Economics* 125(3), 45.

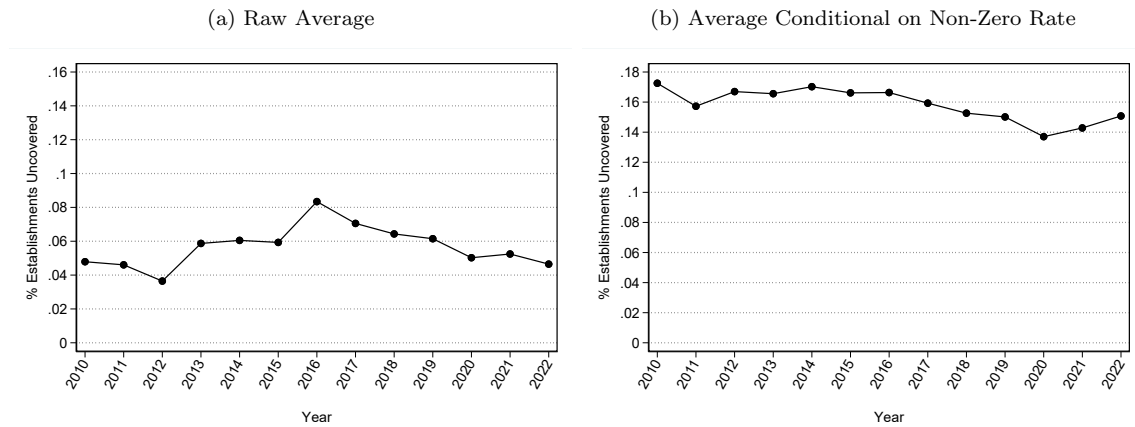
- Fishback, P. V. and S. E. Kantor (1995). Did Workers Pay for the Passage of Workers' Compensation Laws? *Quarterly Journal of Economics* 110(3), 713–742.
- Fishback, P. V. and S. E. Kantor (1996). The Durable Experiment: State Insurance of Workers' Compensation Risk in the Early Twentieth Century. *The Journal of Economic History* 56(4), 809–836.
- Fishback, P. V. and S. E. Kantor (1998). The Adoption of Workers' Compensation in the United States, 1900–1930. *Journal of Law and Economics* 41(2), 305–341.
- Gruber, J. and A. B. Krueger (1991). The Incidence of Mandated Employer-Provided Insurance: Lessons from Workers' Compensation Insurance. *Tax Policy and the Economy* 5, 111–143.
- Gruber, J. and B. C. Madrian (1994). Health Insurance and Job Mobility: The Effects of Public Policy on Job-Lock. *ILR Review* 48(1), 86–102.
- Holmlund, B. (1983). Payroll Taxes and Wage Inflation: The Swedish Experience. *The Scandinavian Journal of Economics* 85(1), 1–15.
- Jinks, L., T. J. Kniesner, J. Leeth, and A. T. Lo Sasso (2020). Opting out of workers' compensation: Non-subscription in Texas and its effects. *Journal of Risk and Uncertainty* 60, 53–76.
- Kaestner, R. and A. Carroll (1997). New Estimates of the Labor Market Effects of Workers' Compensation Insurance. *Southern Economic Journal* 63(3), 635–651.
- Kolstad, J. T. and A. E. Kowalski (2016). Mandate-based health reform and the labor market: Evidence from the Massachusetts reform. *Journal of Health Economics* 47, 81–106.
- Krueger, A. B. and B. D. Meyer (2002). Labor Supply Effects of Social Insurance. In A. J. Auerbach and M. Feldstein (Eds.), *Handbook of Public Economics*, Volume 4, pp. 2327–2392. Elsevier.
- Maclean, J. C. and D. Webber (2022). Government regulation and wages: Evidence from continuing coverage mandates. *Labour Economics* 78, 102236.
- Moore, M. J. and W. K. Viscusi (1990a). *Compensation Mechanisms for Job Risks: Wages, Workers' Compensation, and Product Liability*. Princeton, New Jersey: Princeton University Press.
- Moore, M. J. and W. K. Viscusi (1990b). Have Increases in Workers' Compensation Benefits Paid for Themselves? In *Benefits, Costs, and Cycles in Workers' Compensation*, Huebner International Series on Risk, Insurance, and Economic Security. Norwell, MA: Kluwer.
- Morantz, A. (2010). Opting Out of Workers' Compensation in Texas: A Survey of Large, Multistate Nonsubscribers. In D. P. Kessler (Ed.), *Regulation vs. Litigation: Perspectives from Economics and Law*, pp. 197–238. Chicago, IL: University of Chicago Press.

- Moss, D. A. (2002). *When All Else Fails: Government as the Ultimate Risk Manager*. Cambridge, MA: Harvard University Press.
- Nelson, Jr., W. J. (1990). Workers' Compensation: Coverage, Benefits, and Costs, 1987. *Social Security Bulletin* 53(4), 2–11.
- Richards, M. R. and S. Tello-Trillo (2021). Private coverage mandates, business cycles, and provider treatment intensity. *Health Economics* 30, 1200–1221.
- Rosen, S. (1974). Hedonic Prices and Implicit Markets: Product Differentiation in Pure Competition. *Journal of Political Economy* 82(1), 34–55.
- Rosen, S. (1986). The Theory of Equalizing Differences. In O. C. Ashenfelter and R. Layard (Eds.), *Handbook of Labor Economics*, Volume 1, pp. 641–692. Elsevier.
- Sargent, C., M. A. Bowles, B. J. Sarchet, J. Proudfoot, and Z. Duffly (2020). Recent Developments in Employment and Labor Law. *Tort Trial & Insurance Practice Law Journal* 55(2), 251–272.
- Smith, R. S. (1979). Compensating Wage Differentials and Public Policy: A Review. *Industrial and Labor Relations Review* 32(3), 339–352.
- Sorkin, I. (2018). Ranking Firms Using Revealed Preference. *Quarterly Journal of Economics* 133(3), 1331–1393.
- Spieler, E. A. (2017). (Re)Assessing the Grand Bargain: Compensation for Work Injuries in the United States, 1900–2017. *Rutgers University Law Review* 69, 891–1014.
- Viscusi, W. K. (1979). Job Hazards and Worker Quit Rates: An Analysis of Adaptive Worker Behavior. *International Economic Review* 20(1), 29–58.
- Viscusi, W. K. and T. Gayer (2015). Behavioral Public Choice: The Behavioral Paradox of Government Policy. *Harvard Journal of Law & Public Policy* 38(3), 973–1007.
- Viscusi, W. K. and M. J. Moore (1987). Workers' Compensation: Wage Effects, Benefit Inadequacies, and the Value of Health Losses. *The Review of Economics and Statistics* 69(2), 249–261.
- Weinstein, J. (1967). Big Business and the Origins of Workmen's Compensation. *Labor History* 8(2), 156–174.
- Wissmann, D. (2022). Finally a Smoking Gun? Compensating Differentials and the Introduction of Smoking Bans. *American Economic Journal: Applied Economics* 14(1), 75–106.
- Wriggins, J. B. (2012). Mandates, Markets, and Risk: Auto Insurance and the Affordable Care Act. *Connecticut Insurance Law Journal* 19(2), 275–324.

Chapter II Appendix Tables and Figures

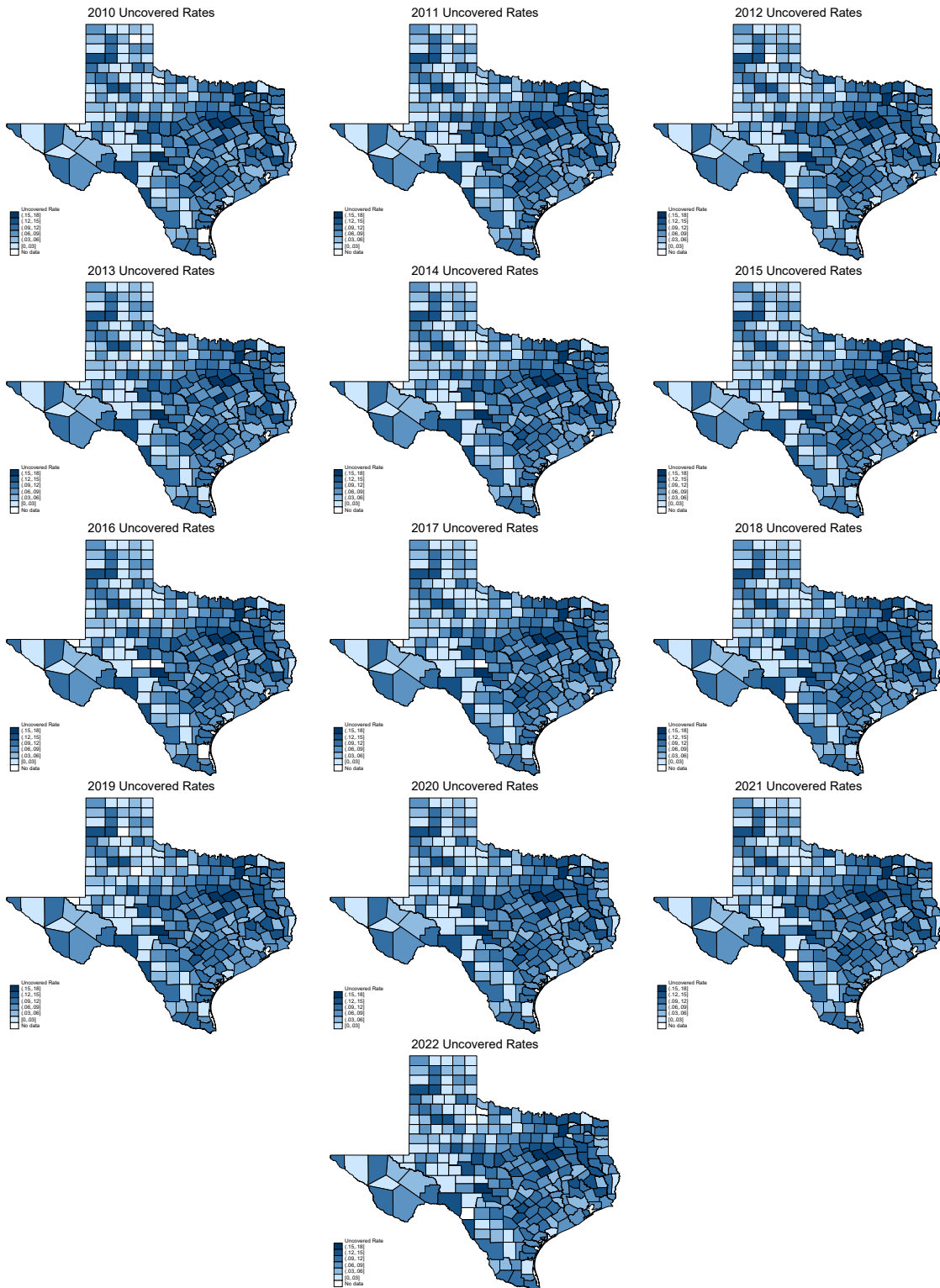
Intended for online-only publication.

Figure A.1: Uncovered Rates in the Data (Unweighted)



Notes: This figure displays uncovered rates over time, for all industry-county observations in panel (a) and among a smaller subset in panel (b) that had a non-zero uncovered rate. Sources: Texas open records requests and QCEW, 2010-2022.

Figure A.2: Uncovered Rates Across Counties by Year



Notes: These maps display uncovered rates by county, broken into several buckets for ease of visualization. The darker the blue, the higher the uncovered rate. Sources: Texas open records requests and QCEW, 2010-2022.

Chapter III - Additional Topics in Workers' Compensation

Tyler Q. Welch

Using the same insurer-level data described in Chapter I and/or the same injury-level data described in Chapter II, I propose several research ideas. Each of these are papers I could work on as a third chapter, which could then eventually be either rolled into one of the other two main chapters upon graduation or pursued as a standalone paper. The ideas below are in no particular order (numbers provided only for ease of reference) and all are attainable with data already in-use or soon to be in-hand (and the focus of my summer work at the Department of Labor). These ideas are all related to workers' compensation, in keeping with the theme of the dissertation and my research agenda moving forward, though I have other papers that are either published (at the *Journal of Health Economics*, focused on the quality of life expectancy) or are complete as working drafts (one on Social Security claiming and Medicaid; the other on a life insurance demand puzzle). It is my preference to keep the theme of the dissertation consistent and to choose one of the new ideas below (or spin off another) with your feedback.

1. Solo: Before mid-2017, all workers' compensation fraud concerns were handled by a specialized department – with criminal referral powers – housed within the state's public option, Texas Mutual Insurance. After concerns about the powers of that fraud unit¹ and pressures from other insurers, the state created a government agency to study fraud and replace this arrangement. Using fraud indicators provided by insurers in the Texas injury data, I would like to study the prevalence of fraud before and after, which not only answers a public-versus-private provision of fraud investigations efficiency-related question but could also

¹Read more here: <https://www.texastribune.org/2017/06/06/private-insurer-will-no-longer-fund-fraud-prosecutions/>.

- lend credence to using NAIC data to study the performance of the public option before and after this change relative to its private market competitors with detailed expense information.
2. Solo: I am interested in using a change in rules associated with compound drug pre-authorizations to deal with prescription costs and opioids in treatment. Texas became concerned about costs and a loophole in their new formulary, which allows drugs made up of ingredients on the formulary individually to be approved together as a compounds. These compound drugs are a concern for medical reasons, as they not approved by the FDA, and for cost reasons, given they are expensive and relevant for potentially addictive pain treatment.
 3. Joint with Ty Leverty: Many public options are exempt from federal taxation. There is no quantification of this tax advantage, however, and this topic is relevant to the broader question of ways in which workers' compensation systems are optimal or sub-optimal for various stakeholders. This project can make use of several state changes in the federal tax exemption (Maine and Utah), along with a few public option privatizations (Arizona and upcoming in Missouri), to study the impact of the federal tax exemption on competition and prices, for example.
 4. Joint with Honglin Li: Workers' compensation is one of the more highly experience-rated lines of insurance in the US. Experience rating is often discussed as a way to solve various informationally-driven market failures. Using reported experience rate modifiers in the injury data from Texas, we propose to study the effect on injuries and claims from 1) changing the size of a claim included in a modifier calculation, and 2) the use of a cutoff in experience rating for small firms. With data available between 2015 and 2021, and these rule changes taking effect in mid-2015 and into 2016, observable policy effectiveness dates allow us to study temporal changes.
 5. Joint with Honglin Li: In 2019, the state of Texas stopped allowing negotiation of experience rate modifiers. This is an important change because some contractors may lose out on bids if their experience rate modifier is above 1 (indicating an injury experience more than

their industry's average). We would like to use this change to think about bargaining power using reported experience rate modifiers and the detailed injury-level data the modifiers are attached to.